MS4 – Supervision of CPS during the engineering and exploitation phases

Esma YAHIA

Esma.yahia@ensam.eu

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Arts et Métiers Sciences et Technologies

MS4 – Supervision of CPS during the engineering - exploitation phases

OBJECTIVES :

Understand the needs and issues of the supervision and monitoring of the processes, humans and the products in the scope of industry 4.0
Define smart monitoring and supervision systems to facilitate decision-making.
Develop competencies in systems engineering, IIoT integration, data acquisition and analysis, and CPS design.

STRUCTURE :

S4.1 : Overview of the CPS supervision and monitoring issues

- Industry 4.0's contribution to systems supervision and decision support — 2h (EY)
- Introduction to IIoT architectures — 1h (EY)

S4.2 : Application : Project on monitoring system production

- Project Kick-off and Expectations — 2h (EY)
  - Definition of system objectives and success criteria
  - Identification of critical monitoring points, sensors, and data sources
- Project Requirements and Preparation Phase — 2h (EY)
  - Functional requirements for monitoring systems
  - System architecture design (hardware/software integration)
  - Data acquisition, IIoT connectivity, and dashboard setup
- Project Work Sessions — 14h (EY)
  - Implementation of monitoring prototypes
  - Real-time data collection, analysis, and visualization
  - Testing, validation, and iterative improvement

PRACTICAL ISSUES :

Team : E.Yahia
Evaluation : 1h exam + 2h oral

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Design of monitoring and decision-support systems

Systems engineering

Decision support system

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criteria grid

Innovgames project

incident resolution project

Evaluation: ORAL

Scenario-based exam

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Esma YAHIA, PHD

- Short CV

- Associate professor at ENSAM –Campus Aix-en-Provence since 2012
- Invited professor at Polytechnique Monteral 2023-2024

- Phd thesis on semantic interoperability evaluation with an application to product systems information systems

- Master's thesis in Information systems design

- Industrial Engineer

Contact: esma.yahia@ensam.eu

Esma YAHIA, PHD

• Phd thesis supervision

- Ghofrane Ben Yarrou (Industrial project SYNCRONE)

  Digital continuity within information systems based on an interoperability approach combining model-driven engineering and learning algorithms.

- Zhyiu Zheng (Chaire BIM)
  Digital twin framework for data interoperability and continuity between the design phase (and the operation phase (BMS).
- Jean René CAMARA (Capgemini Engineering)
  Knowledge management in design and prototyping processes.
- Quentin Brilhault (Thales Alenia Space and CNES)
  Digital continuity based on learning transformation models: a model-driven interoperabilit approach.
- Lise Kim (Capgemini)
  Creative implementation of data science in a distributed, modular, and heterogeneous digi engineering environment.
- Kenza Amzil (doctoral scholarship)
  Realization of a supervision and multi-view monitoring system in real time for decision support.
- Widad Essoufi (FUI)
  Modeling and process mining to assist decision-making in the context of system design and supervision.



Allez sur wooclap.com

Entrez le code d'événement dans le bandeau supérieur

Code d'événement

ZRMSYT

Envoyez @ZRMSYT

au 06 44 60 96 62

Copier le lien de participation

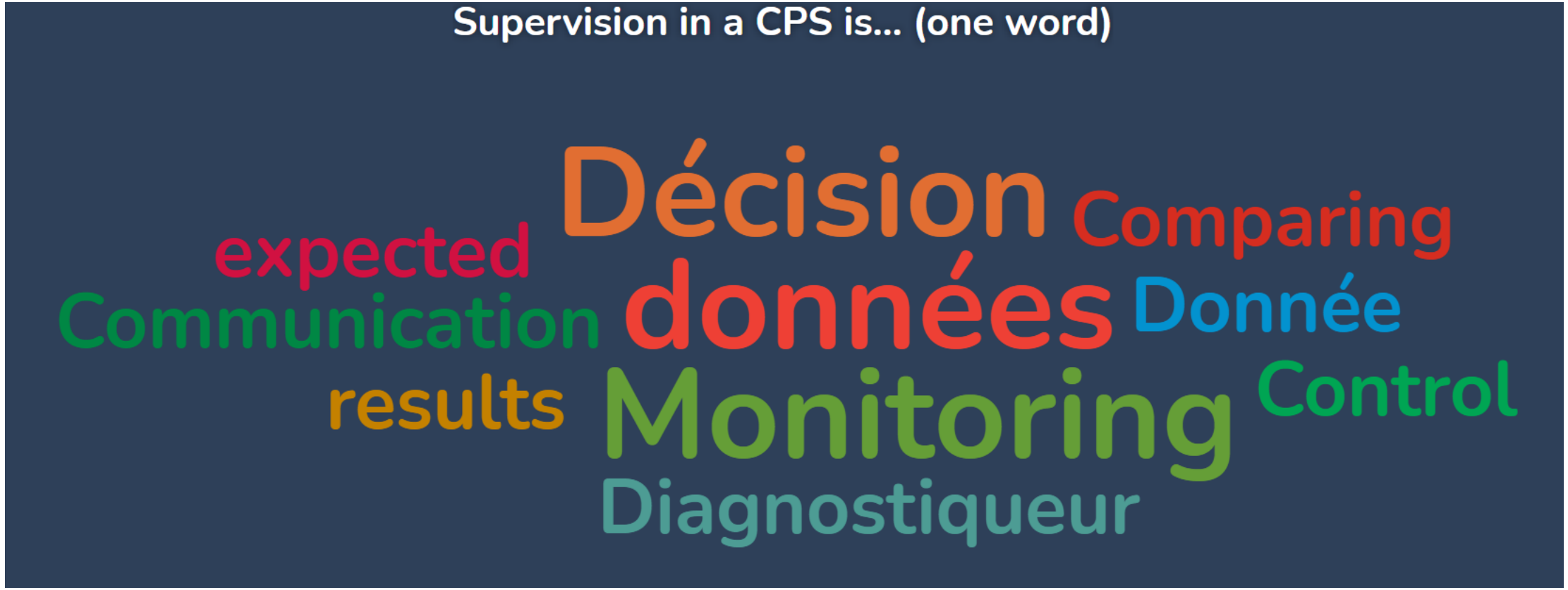
1

participant

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1. Supervision in a CPS is… (one word)
2. What should supervision help you decide or do that you could not do without it?

What is supervision in the context of CPS?



What should supervision help you decide or do that you could not do without it?



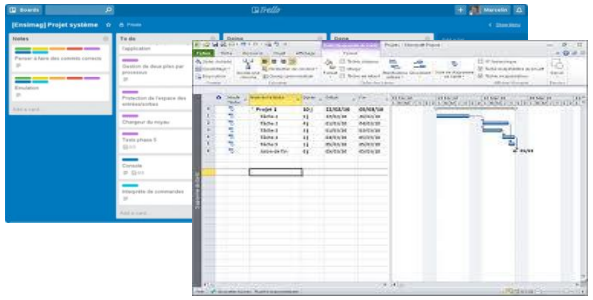
Supervision …





Supervision of train traffic

Supervision of information systems, networks, servers…





Supervision of engineering projects

Supervision of large-scale systems

Monitoring of physiological indicators



Supervision of the gas network

Supervision of production systems





Supervision of dairy product manufacturing

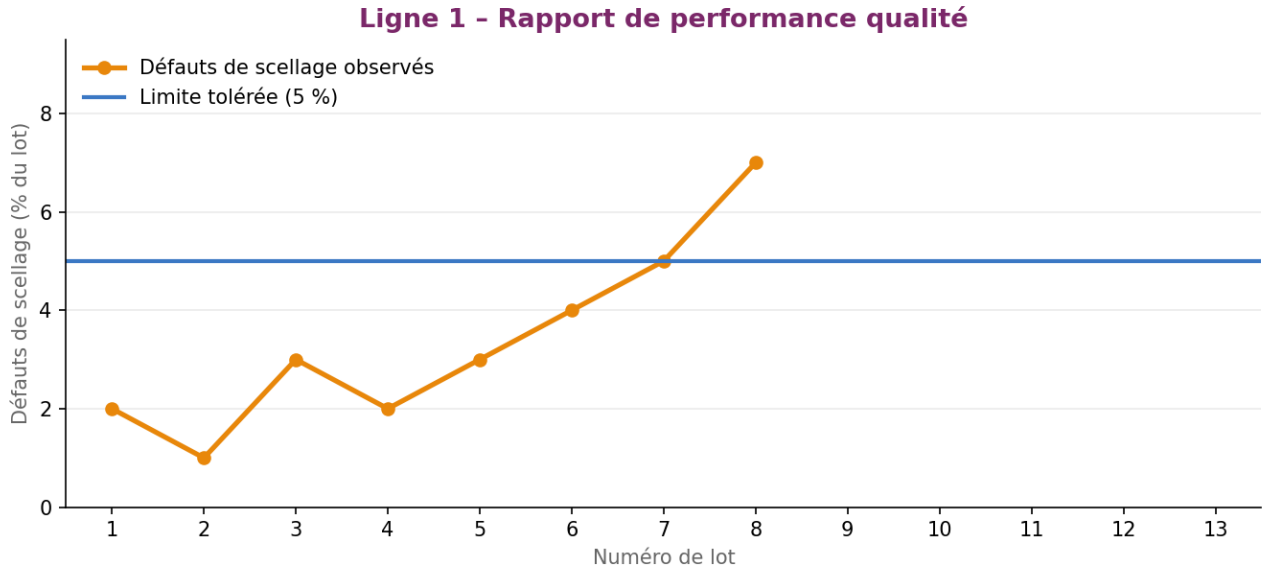


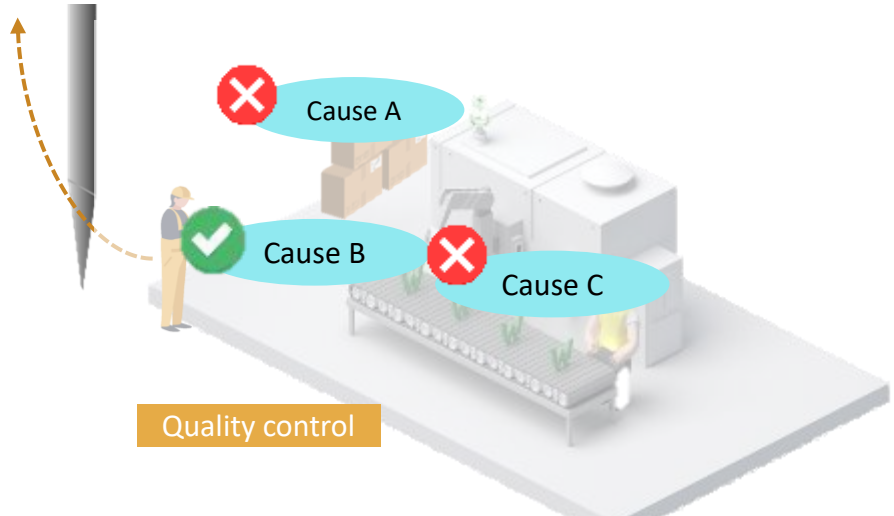
Supervision of car assembly

Arts Sciences et Technologies et Métiers

What is the link between supervision and decision support?

- Mise en situation: Production System – Problem Solving





What is the link between supervision and decision support?

- **Mise en situation: Production System – Problem Solving**

FICHE 1 · AUTOMATE (journal machine, PLC/SCADA)

Votre source : l'automate de la scelleuse enregistre en continu ses mesures. Personne ne consulte ce journal sans alarme ni demande particulière.

Références de la fiche procédé : consigne de température de soudure 180 °C, **tolérance produit ± 5 °C** ; pression de scellage 4,0 bar ; cadence nominale 120 pots/min.

Lot	T soudure moyenne (°C)	Pression (bar)	Cadence (pots/min)
1	180	4,0	120
2	181	4,0	120
3	179	4,1	119
4	180	4,0	120
5	181	4,0	121
6	178	4,0	120
7	175	4,1	120
8	171	4,0	119
9	170	4,0	120
10	169	4,0	120
11	169	4,1	120

Journal d'alarmes

- Seuil d'alarme « température basse » : **170 °C** (réglage d'origine, jamais modifié).
- Lot 10, 10 h 42 : alarme « T basse » déclenchée. 10 h 43 : acquittée par l'opérateur.
- Aucune autre alarme sur la période.

FICHE 2 · OPÉRATEUR (fiche papier de fin de poste)

Votre source : les fiches manuscrites de l'opérateur de la ligne, Marc D. Elles sont remises au chef de ligne le lendemain, puis ressaisies dans un tableur le vendredi.

Lot	Notes manuscrites
1 à 5	Poste du matin. « RAS. »
6	« Sonde et résistance changées hier par la maintenance. RAS. »
7	« Quelques pots mal scellés en sortie. Pas plus que d'habitude, je trie à la main. »
8	« Beaucoup plus de pots à trier. Fin de poste en retard. »
9	« Karim (intérimaire) en renfort à partir d'aujourd'hui, formé 1 h sur la ligne. Même rebuts qu'hier. »
10	« Alarme T basse à 10 h 42. Acquittée. Pas de consigne écrite pour ce cas. »
11	« Toujours autant de rebuts. »

FICHE 3 · MES / ERP (ordres de fabrication, approvisionnements, planning)

Votre source : le système de gestion de production. Il n'a aucun lien automatique avec le journal de l'automate.

Lot	Film d'operculage utilisé	Équipe (planning RH)
1 à 6	Film **F-2207**, fournisseur Alpi-Pack	Équipe titulaire (Marc D.)
7 et 8	Film **F-2231**, nouveau lot Alpi-Pack, réceptionné le lundi	Équipe titulaire (Marc D.)
9 à 11	Film F-2231	Équipe titulaire + 1 intérimaire en renfort

Informations complémentaires

- Le certificat de conformité fournisseur du film F-2231 est **conforme**.
- La **ligne 2** utilise le même film F-2231 depuis son lot L2-04, soit la même période que le lot 7 de la ligne 1. Défauts de scellage de la ligne 2 sur les lots L2-04 à L2-08 : **1 %, 2 %, 1 %, 2 %, 1 %**.
- Vous ne connaissez ni les températures de la scelleuse, ni les interventions de maintenance.

FICHE 4 · MAINTENANCE (GMAO)

Votre source : la gestion de maintenance assistée par ordinateur : bons de travail, stock de pièces.

Bon de travail BT-4471, fin du lot 5

- Maintenance préventive planifiée : remplacement de la **sonde de température** et de la **résistance chauffante** de la barre de scellage.
- Note du magasin : la référence habituelle **RC-120** était en rupture. Elle a été remplacée par la référence **RC-90**, jugée « compatible » par le fournisseur, de **puissance inférieure de 25 %**.
- Contrôle après intervention : « fonctionnement OK » (démarrage à froid, contrôle visuel).
- Le bon de travail **ne prévoit pas** de relevé de température indépendant ni de recalibrage.

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What is the link between supervision and decision support?

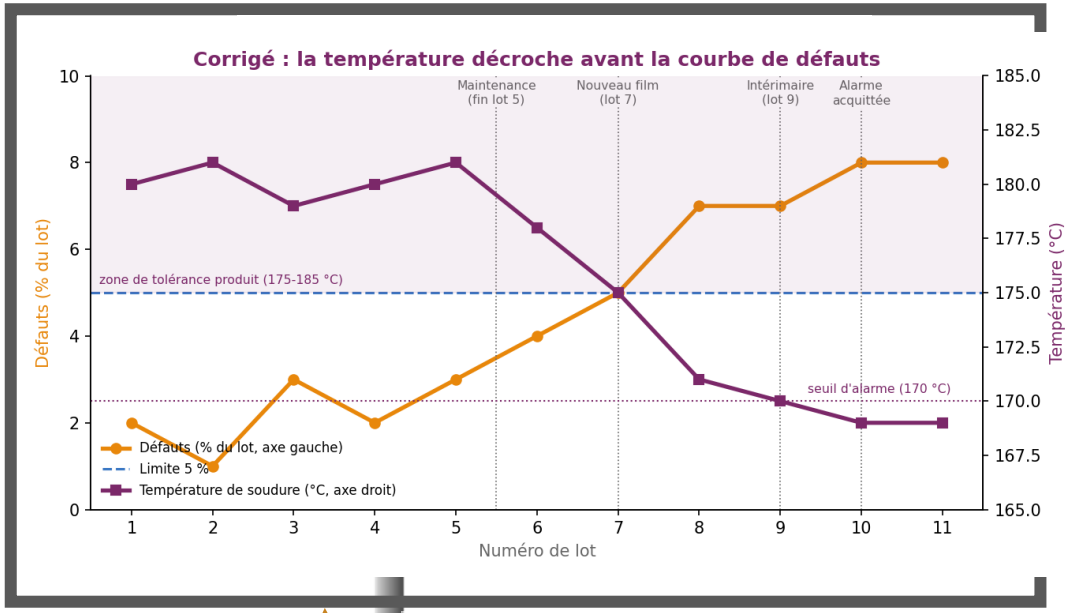
- **Mise en situation: Production System – Problem Solving**



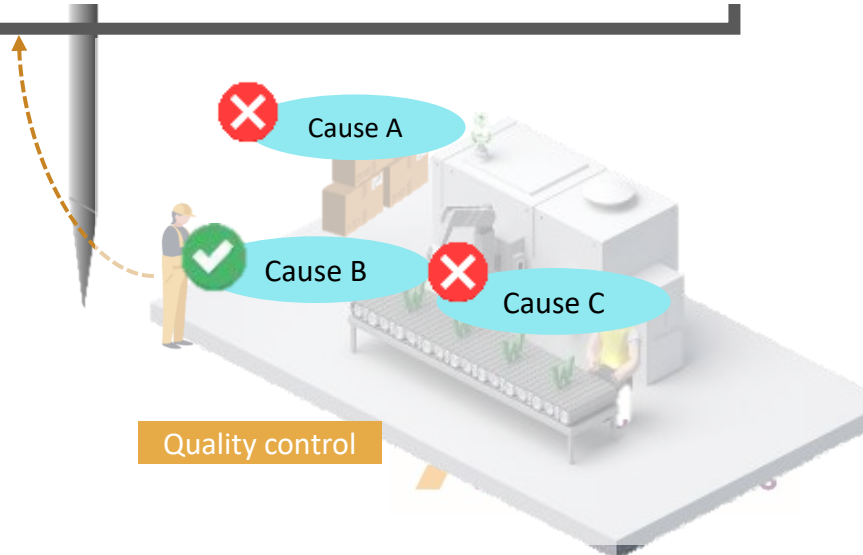
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What is the link between supervision and decision support?

- Context: Production System – Problem Solving



Supervision provides real-time awareness, while decision support transforms that awareness into informed, effective action



Expectations: Supervision… to manage…

- Monitoring and control to support decision-making

Monitoring involves observing and providing transparent information on the system's health by tracking **performance indicators** and **detecting malfunctions**.



- **Control** means acting on the execution of processes and/or the monitored systems, such as through parameter adjustments, calculations, rescheduling, optimization, redefinition, or switching from one control law to another based on the current state of the process and/or system

Managing through decision-making to:

- Resolve and reduce malfunctions
- Improve the operational performance of the organization/project
- Anticipate future deviations
- According to a given strategy

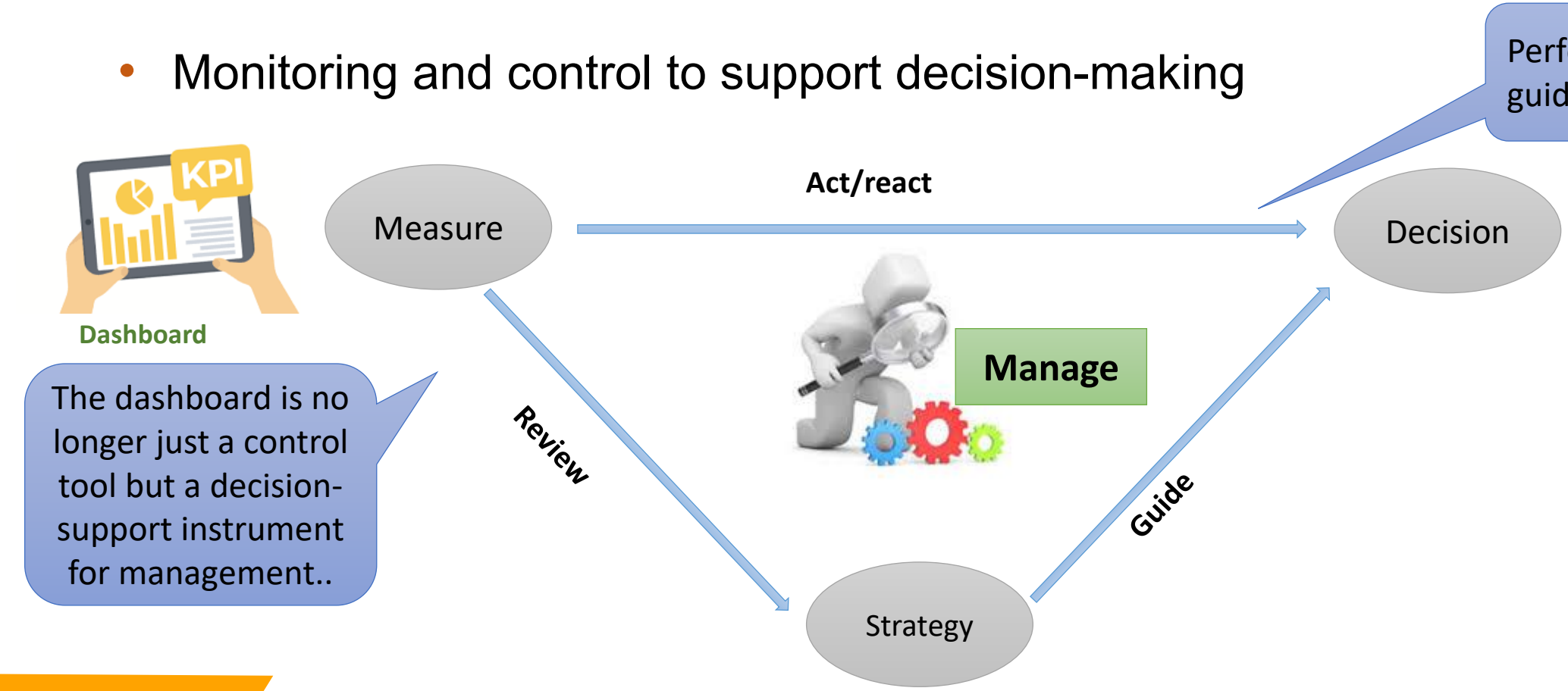
Mirdamadi, 2009

Expectations: Supervision… to manage…

- Monitoring and control to support decision-making

Dashboard

The dashboard is no longer just a control tool but a decision-support instrument for management..



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Expectations: Supervision… to manage…

- Monitoring and control to support decision-making



Dashboard

Measure

Act/react

Decision



The airplane metaphor



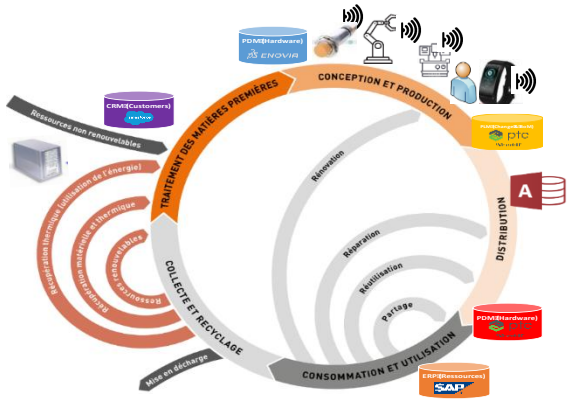
Supervision is like flying an airplane:
Monitoring instruments = checking the plane's health and performance indicators
Correcting deviations = adjusting the flight path if the plane drifts from its intended course
Decision-making = changing altitude, speed, or route in response to weather, traffic, or other external events
Target adjustment = modifying the flight plan if priorities or conditions change

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Supervision… with a focus to the knowledge

- Monitoring and control to support decision





- **Different types of data:**
  - Sensors, information systems, automated machines, documents (digitalised and not)
- **Stored on various media:**
  - Databases, data lakes, etc.

- **Data in silos :**
  - Generated by software (ERP, MES, etc.) for specific business needs
  - Managed through different information systems
  - Often stored in PDF documents and Excel files
- **Increasing volume of data**
  - 300,000 parts for an airplane
  - Sensors (IoT),
  - Internet+1.7 MB/second/person in 2020 [IDC]
  - 450 million GB
- Data are **redundant, inconsistent, poorly interoperable, and not reliable for decision-making.**





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Supervision… with a focus to the KPI

- Monitoring and control to support decision-making

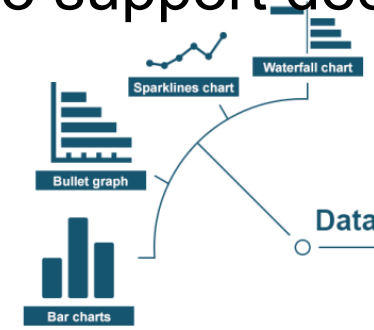


Tableau de bord

Measure



Obeya



Data Visualization

Key Performance Indicators (KPIs)

It shows how good your business is performing. They are used to achieve better results, understand your business and make effective, strategic decisions.

KPI

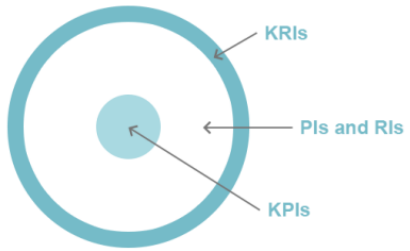
4 Types of Performance Measures

Key result indicators (KRIs): Give an overview of past performance and are ideal for the board as they communicate how management has done in a CSF or from a balanced scorecard perspective.

Performance indicators (PIs): Tell staff and management what to do.

Result indicators (RIs): Tell staff what they have done.

KPIs (KPIs): Tell staff and management what to do to increase performance dramatically.



KPI ≠ Metric

Not All Metrics are KPIs

Do not assume that **KPI** and metrics are the same. **KPI** is a metric, but metric is not necessarily a **KPI**. What makes a metric **KPI**?

- **KPIs** follow company's goals, it leads to action;
- They are decided by management;
- **KPI** has a meaning on all organizational levels;
- It provides context and is based on legitimate data;
- **KPI** is simple and easy to understand.

Leading and Lagging Indicators

Lagging KPIs are focused on the past.

Leading KPIs are focused on the future.



S.M.A.R.T.

Measure What Matters

This method for defining criteria for your effective **KPIs**. **S.M.A.R.T.** is an abbreviation (mnemonic acronym), giving criteria to guide in the setting of objectives conforming to the following criteria:

- **S – Specific** purpose for the business, target a specific area for improvement.
- **M – Measurable**. Quantify or at least suggest an indicator of progress. KPIs should allow you to set measurable goals for improving core business activities.
- **A – Attainable** (or Achievable) state what results can realistically be achieved, given available resources.
- **R – Relevant** to the success of the organization
- **T – Timed**. The value or outcomes are shown for a predefined and relevant period.

Supervision… with a focus on decision support…

Caracterisation od decision making situation

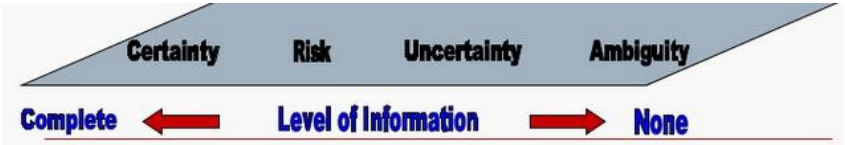
- **Programmed situation:** when the decision responds to a known situation, so it can be identified from a list of past decisions
- **Non-programmed situation:** when the decision responds to a unique, unstructured, or poorly defined situation

Making a decision to solve a problem:

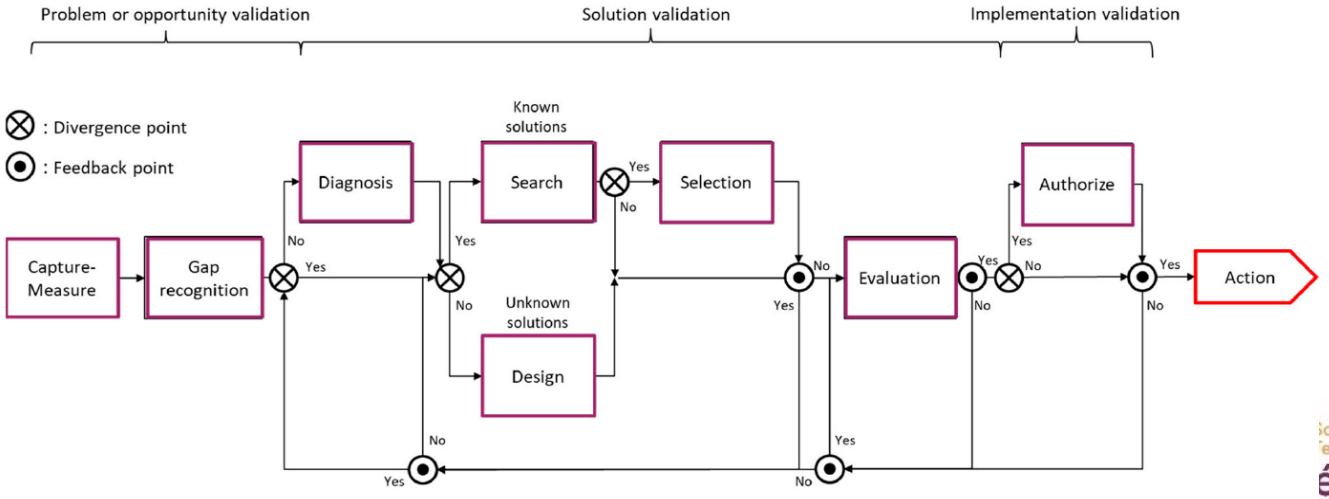
- **Opportunity: a new, more favorable situation**
- **Deviation: a situation drifting from a reference point (the gap)**

Decision

Decsion making condition

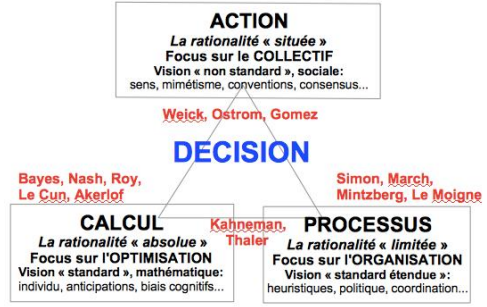


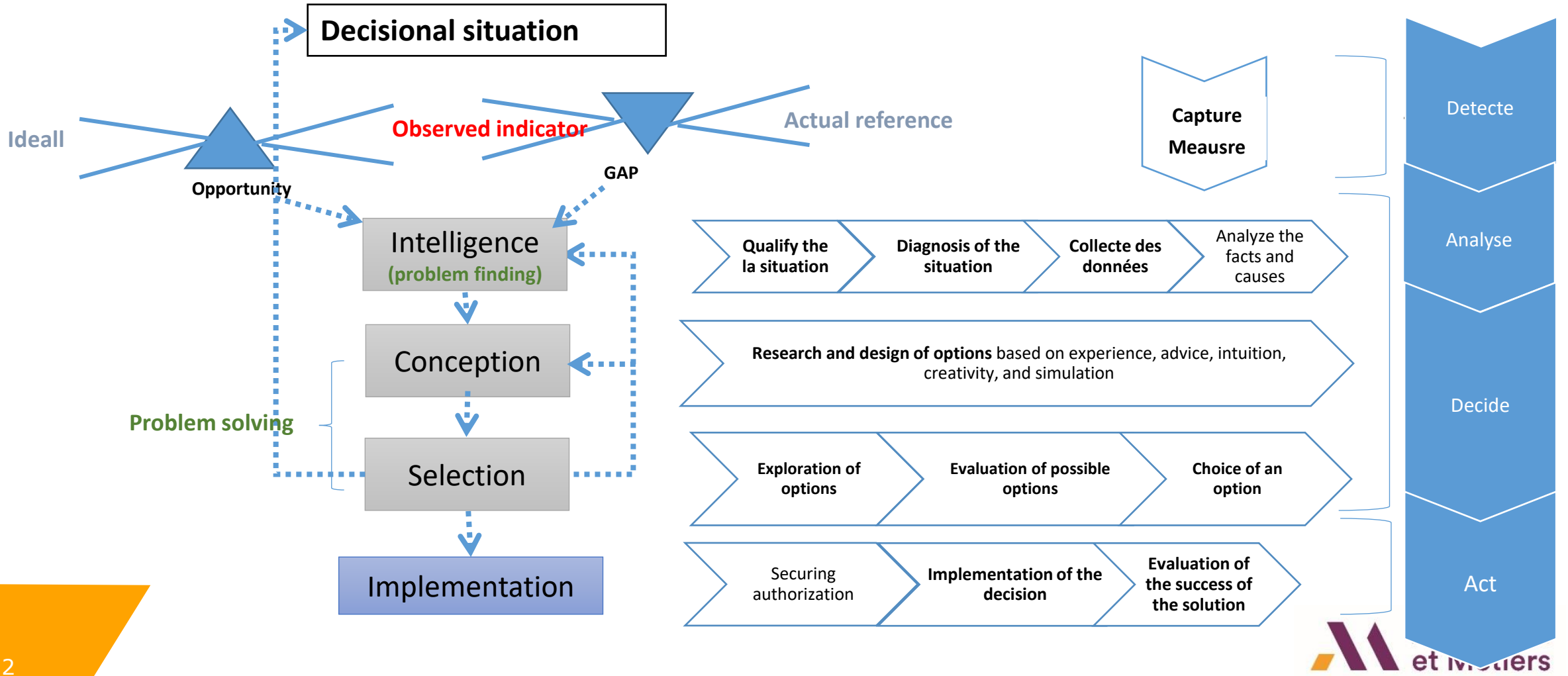
Decision making process



(Sterman, 2000).

How to make a decision?....





H. Simon model and the porpsed models of H. Mintzberg et J-L Le Moigne

Key elements for supervision to work effectively



Measure

Alarme

Charts

Graphs

KPI

Data visualisation

Decision aid

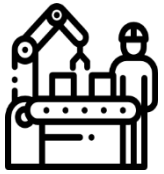
Decision

data acquisition from the field

DATA BASES…..

instructions and actions

Data acquisition









Detecte

Analyse

Decide

Act

Key elements for supervision to work effectively

Data Visualisation

Alarme | Charts | Graphs | KPI

Data processing

Data cleaning → Data reduction → Data analysis → Data mining

Data storage

Structured data | Unstructured data

Data collection

HUB IOT | Machines Log | Product Log | SI Log

Data Source

MES/ERP/...

Suppliers | Manufacturing | Assembly | Design | Recycle | Maintenance | Utilization | Sale | Logistics

Data Analytics



Detecte

Analyse

Decide

Act

Key elements for supervision to work effectively

Technologies

Watson IBM, IA, Simulation, Digital twin

Managment visuel, Node-red Dashboard, Grafana, Tableau, i-obeya, …

Prom, data mining, NLP, OCR ….

Cloud, Data lake, Mongo DB, My SQL , NOSQL

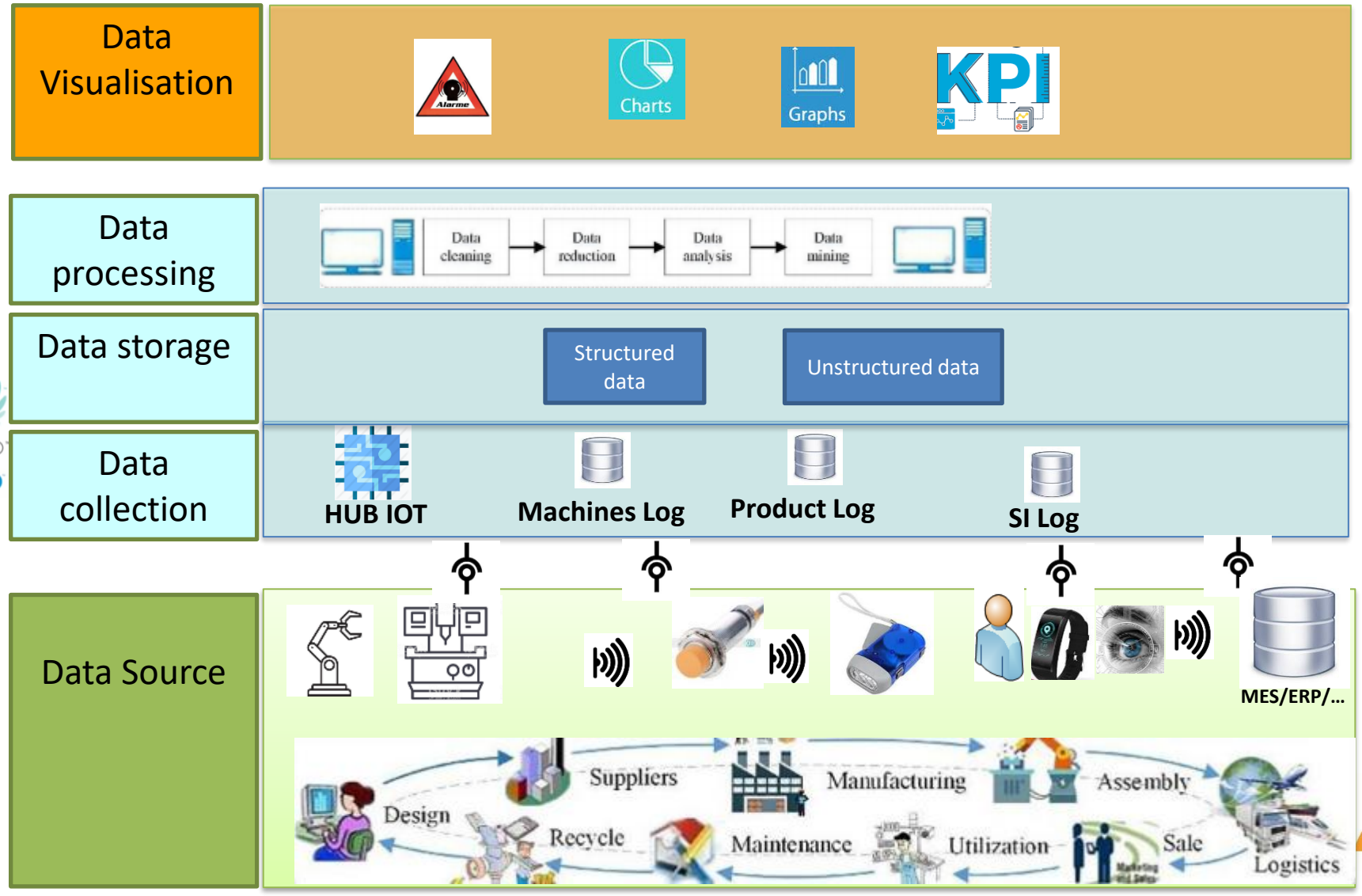
IOT process engines

kura, Node-RED, FLOGO, eclipse, Apache Camel, IBM, TIBCO

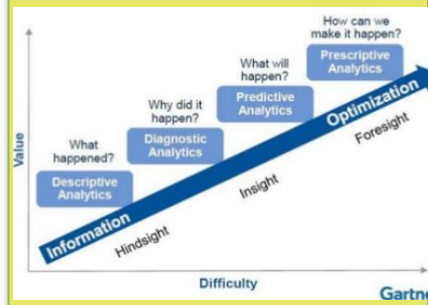
Dataflow piplines:
ETL (Extract/Transform/Load)
ESB (Enterprise Service Bus)
API (Application programming interface)

Différents protocoles de communication: Bluetooth, Lora, Zigbee, TCP/IP, SIGFOX, MQTT, MODBUS…..

Différents standards de communication: OPCUA (OPC Unified Architecture)



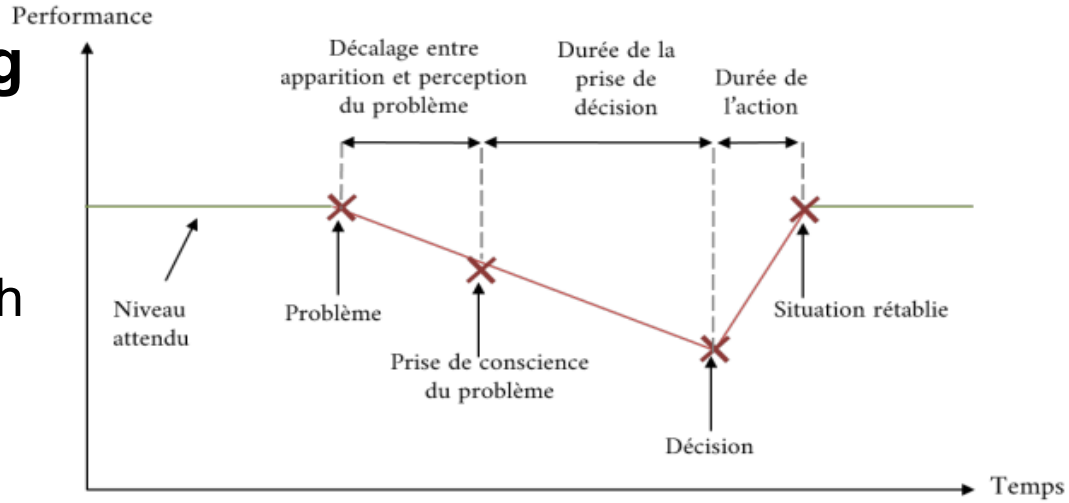
Data Analytics



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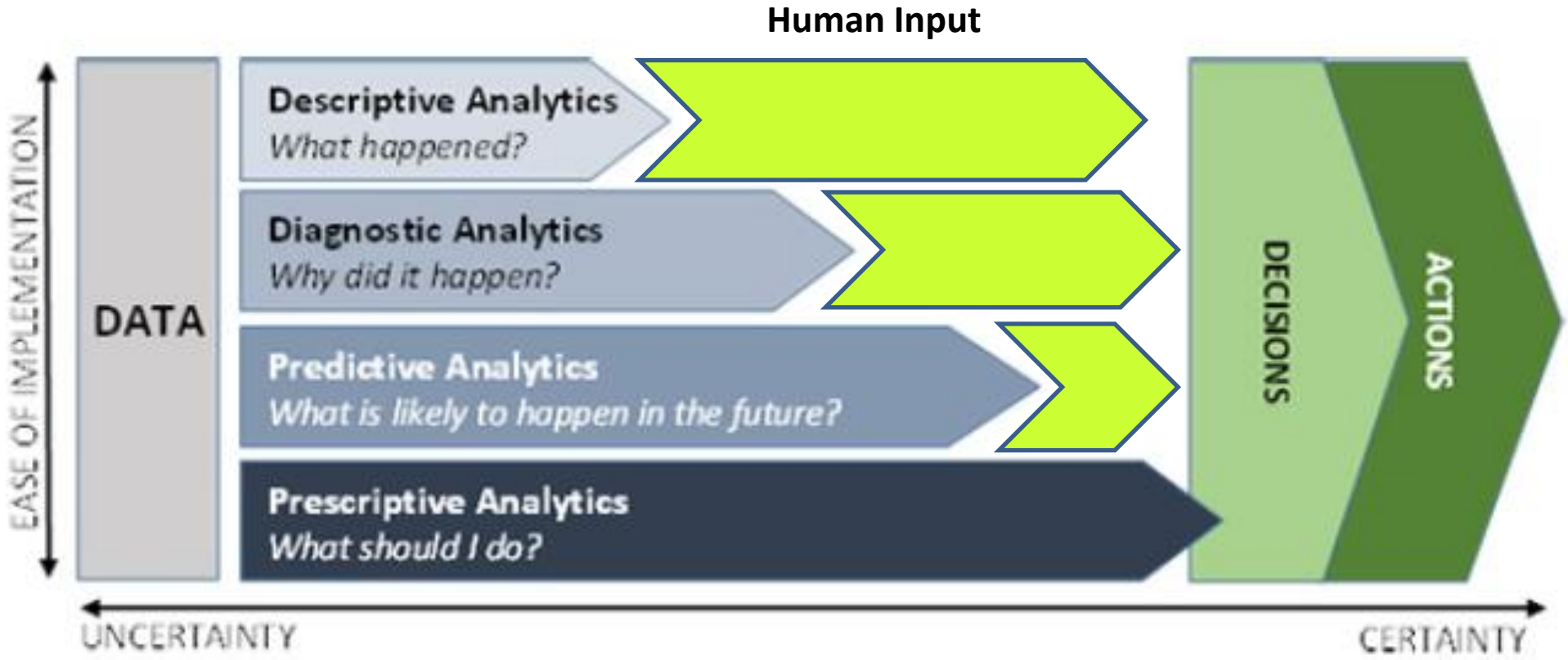
Key Questions about Decision-Making

- **How to make the best decision?**
- **How to accelerate the decision-making process?**
  - What are the facilitators/enablers?
  - How to design a decision support system with consideration of:
  - Human-centered design,
    - Data-driven manufacturing,
    - Digital continuity,
    - User empowerment,
    - Sustainability



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Turning data into actionable decisions for better business outcomes



Gartner 2014

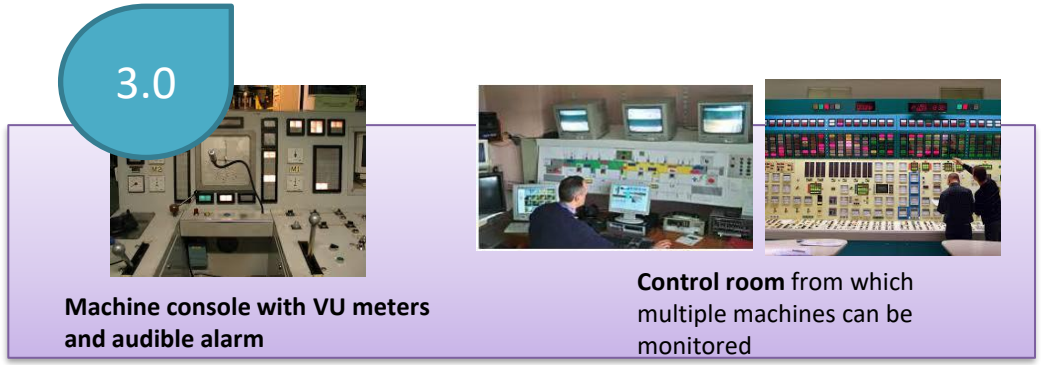
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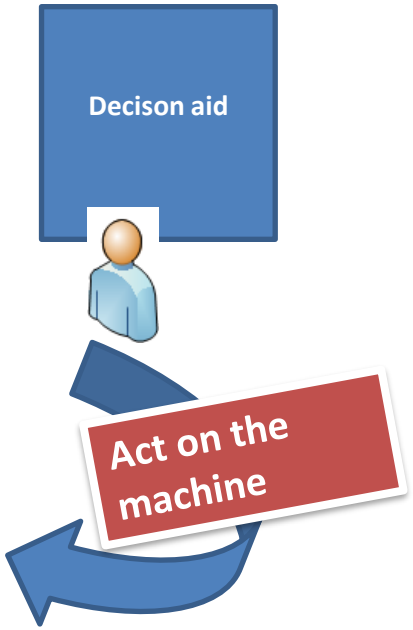
To go further...

- Supervision across different eras
- Enablers / Facilitators
- Examples

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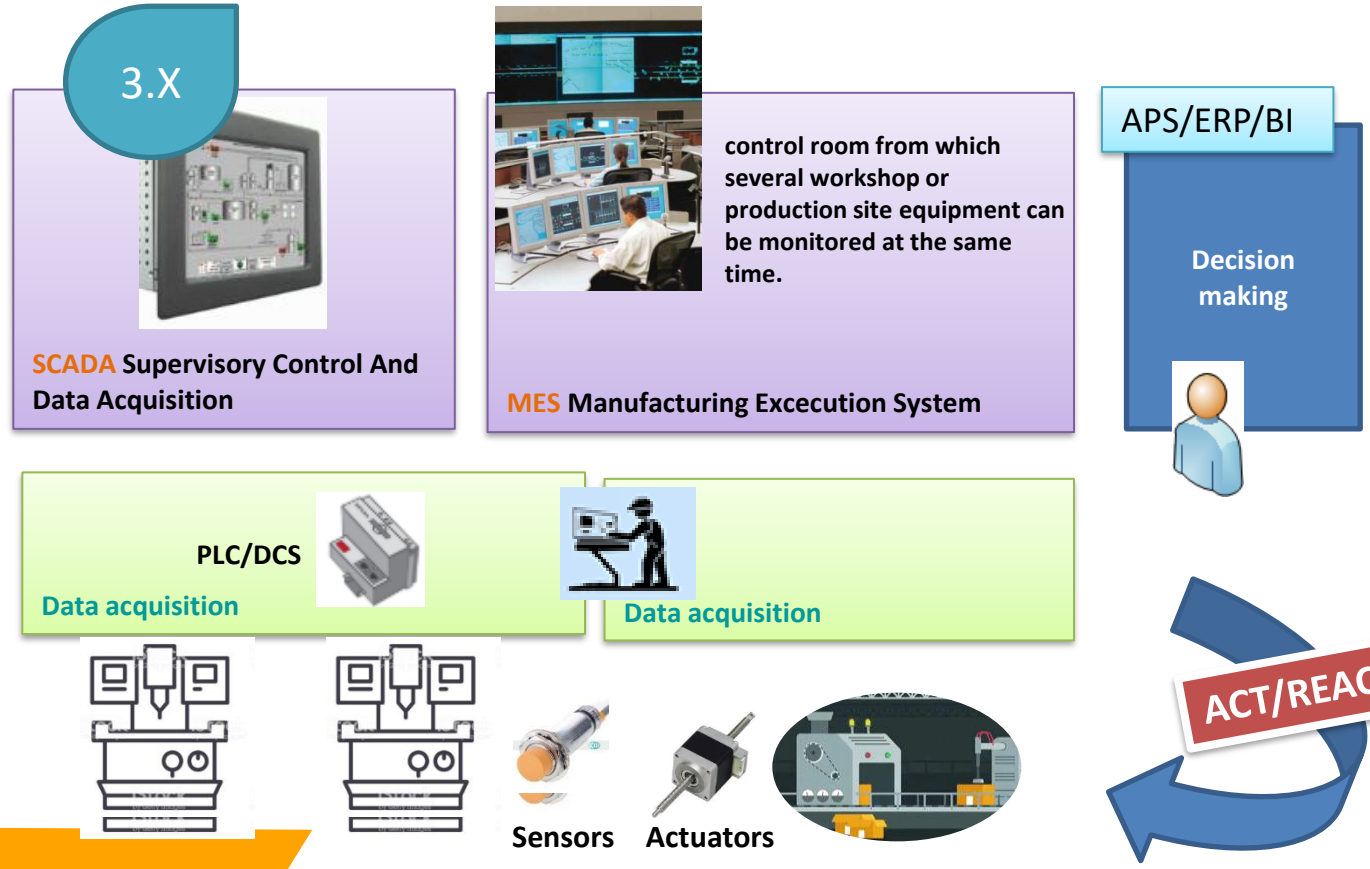
Supervision… in the era of Industry 3.0



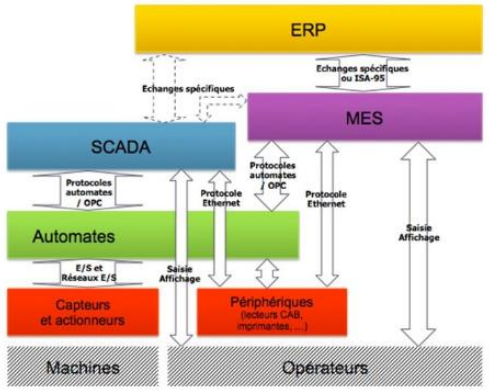


- Supervision with a focus on production resources
- Decisions are made by the operator

Supervision... in the era of Industry 3.X

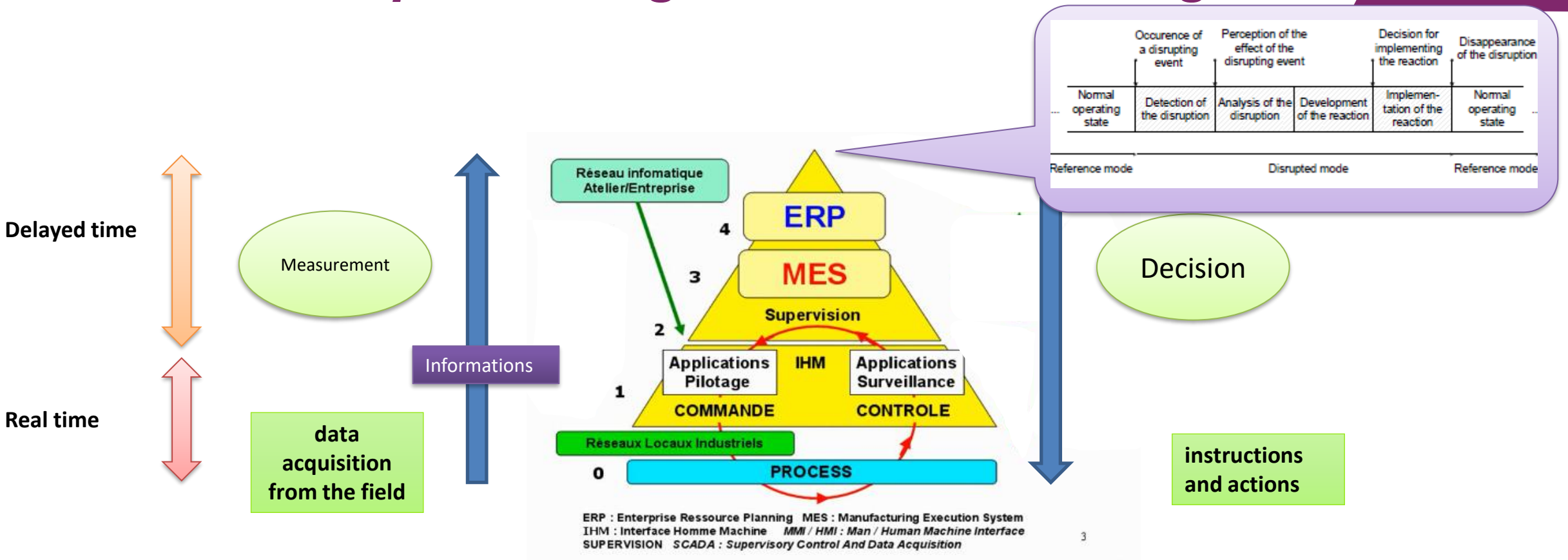


- Supervision... focus on production resources and information systems.



- Decision-making is based on information provided by information systems and will be carried out by the operator and automated systems (opening/closing of valves, for example).

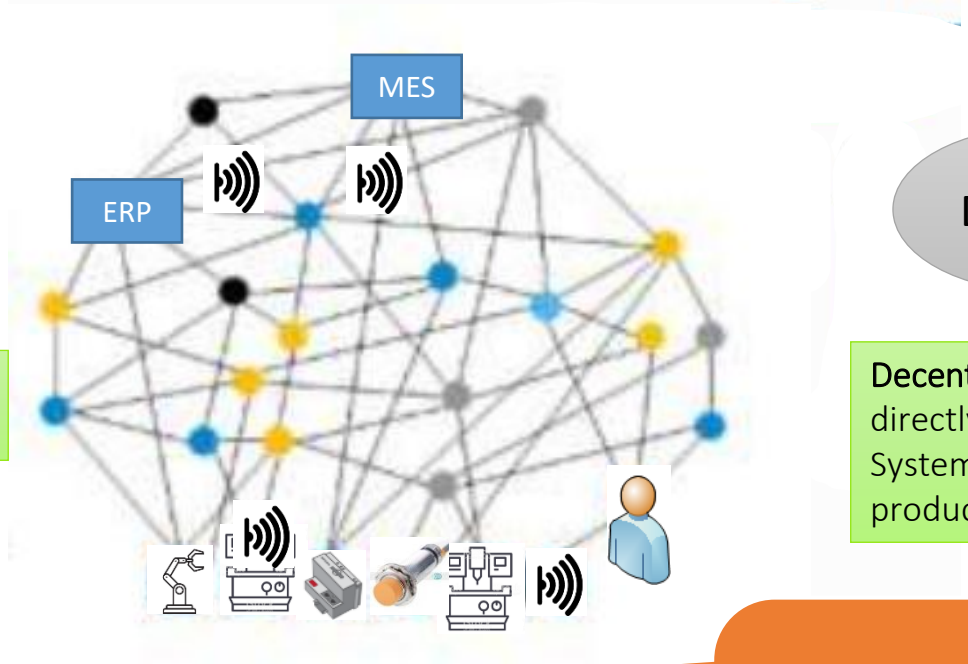
CIM *computer-integrated manufacturing*



The CIM pyramid in the era of Industry 4.0

Measure

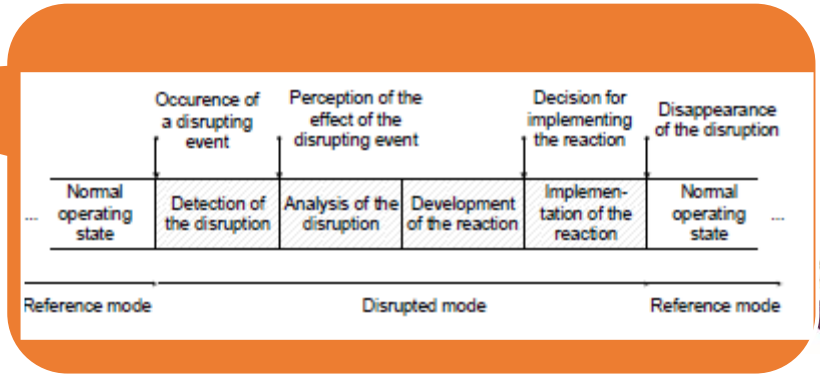
Real-time communication of information for monitoring



CPS+ IIOT

Decision

Decentralized decision-making directly on CPS (Cyber-Physical Systems) as close as possible to production



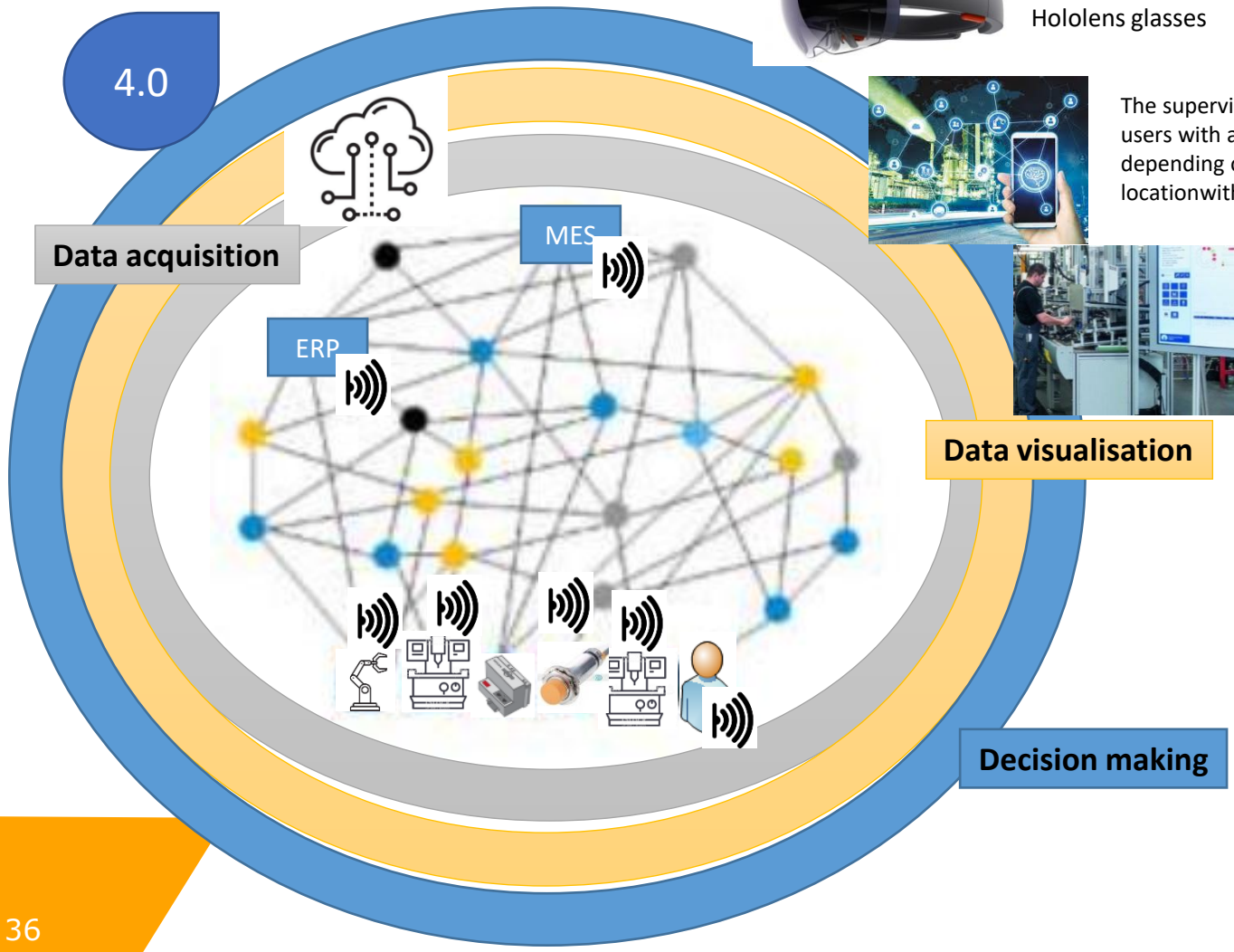
Supervision in the era of Industry 4.0



The operator can view alerts directly on Hololens glasses

The supervision interfacecan provide users with a 360° viewof their processes, depending ontheir role and their locationwithin the company.

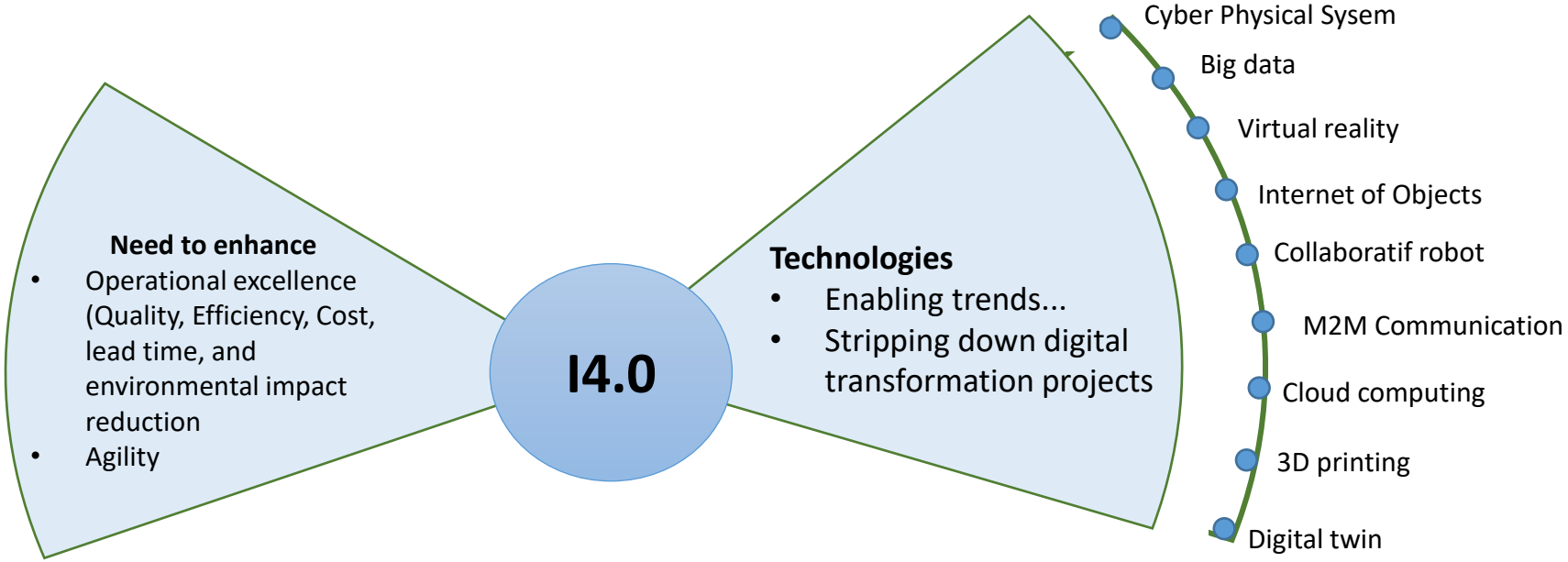


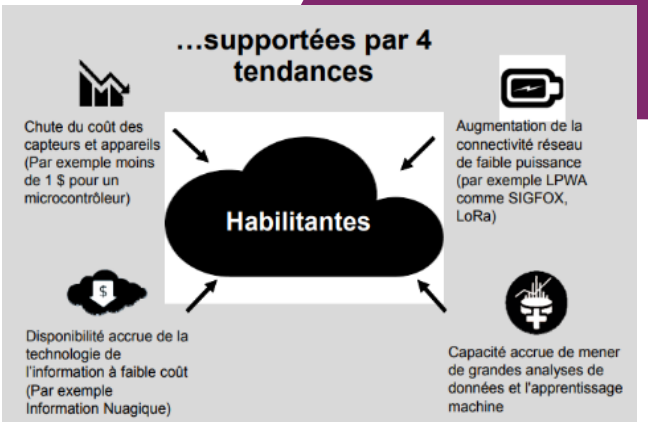


- The supervision interface can provide users with a 360° view of their processes, based on their role and their location within the company.
- Decision-making is therefore decentralized/distributed and rapid, as alerts are provided in real time..

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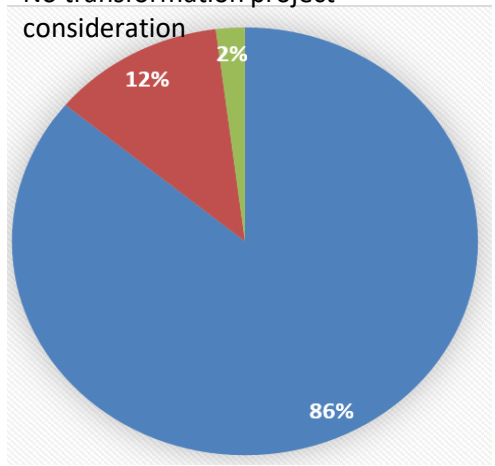
Enablers for fast and effective decision-making





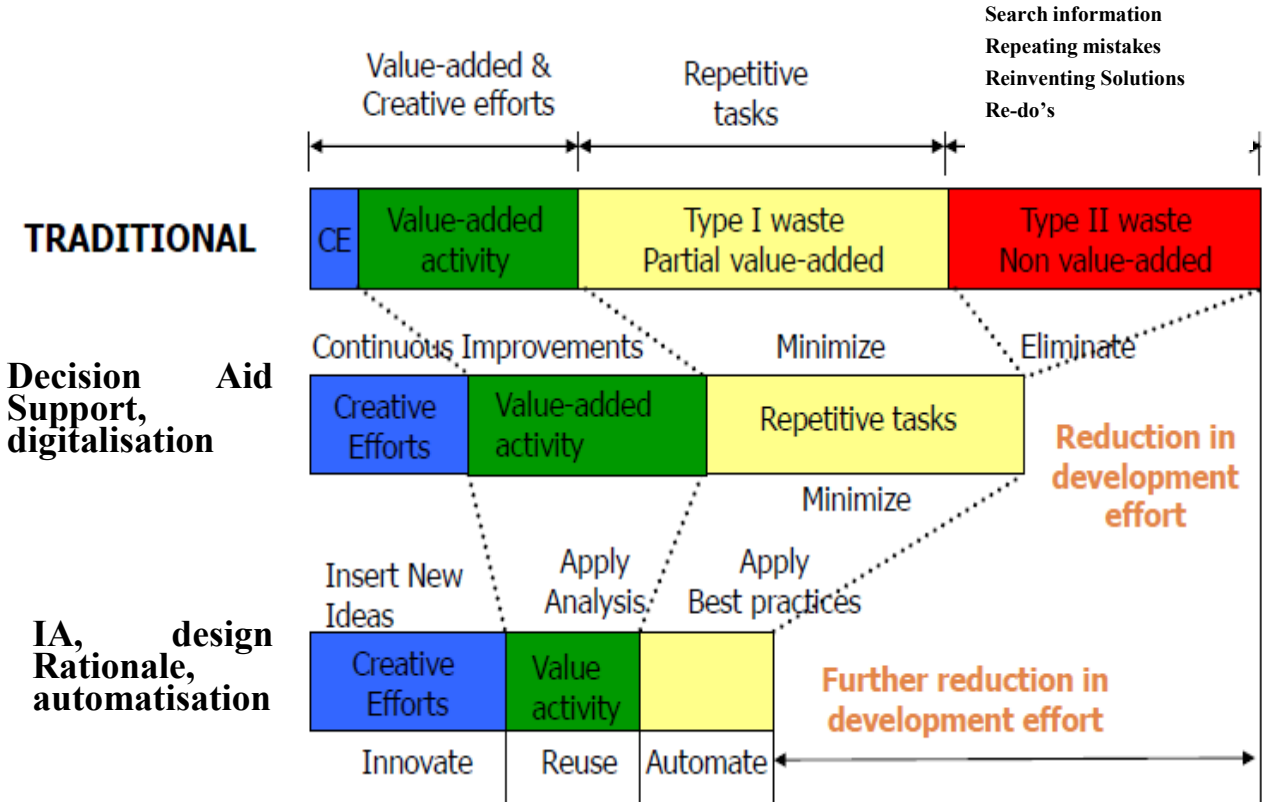
[CEFRIO ,19]

- Transformation project already launched
- Transformation project Under consideration
- No transformation project consideration



Adapted from (Marie & Poindextre, 2020

Industry 4.0…expected performance vs. ROI



Productivity gains: higher throughput and efficiency, eliminating non-value-adding activities.

Failure prevention: maximizing OEE, avoiding re-work, scrap, outages, and poor product quality.

Flexibility: hiding complexity, low configuration or reconfiguration effort, plug and produce, and avoiding technology gaps.

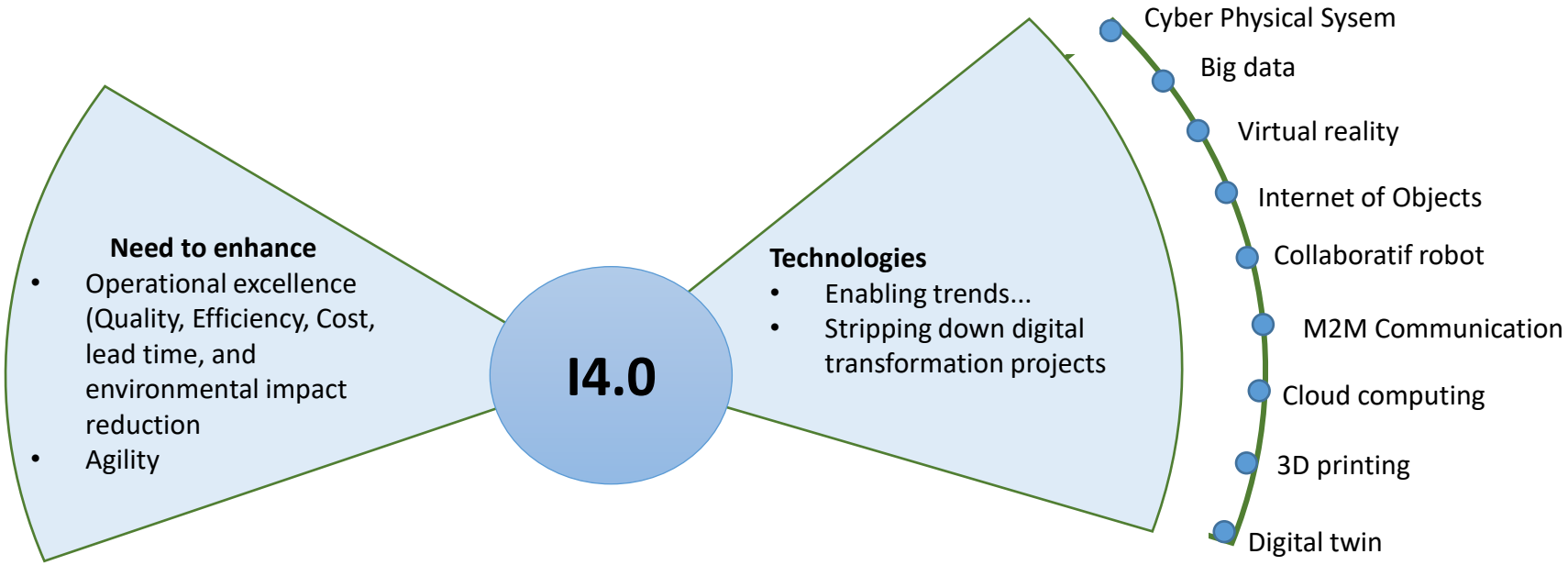
Distributed/decentralized decision making: faster decision making, more actionable decisions

Agility, collaboration, real time

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Enablers for fast and effective decision-making

Challenges

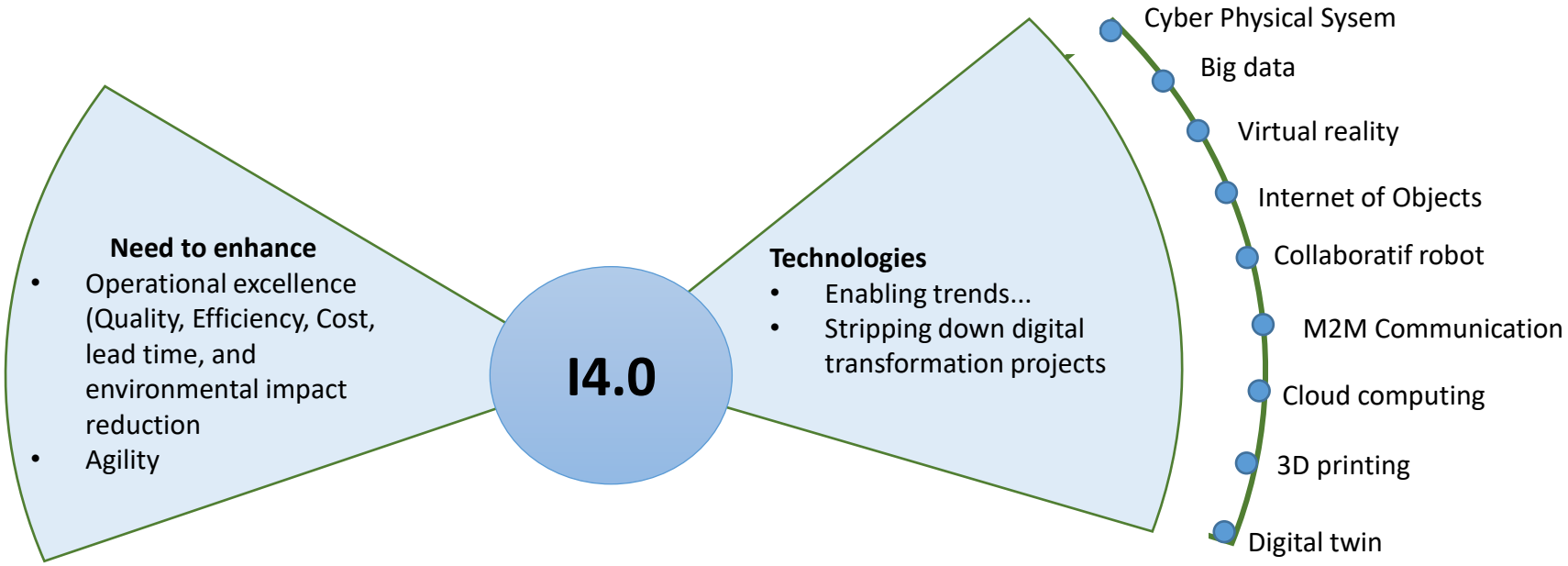


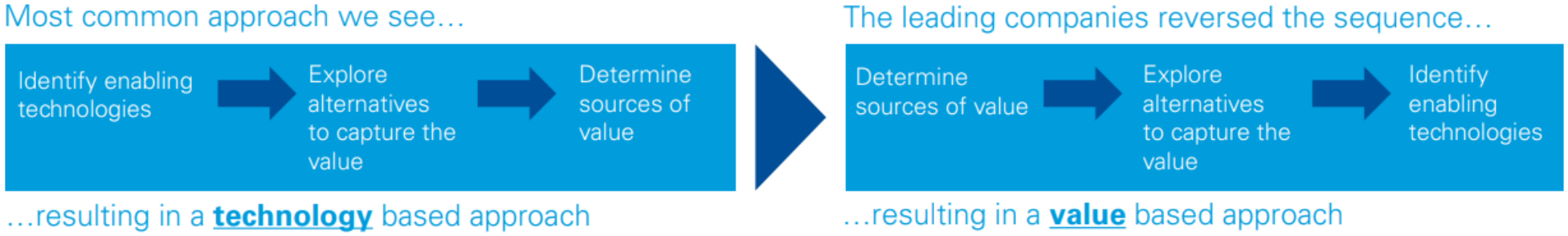
- Industry 4.0 initiatives with different maturity levels
- Lack of alignment between initiatives and with the overall strategy
- Resistance to change and lack of buy-in from company stakeholders (skills, clarity of objectives, layoffs, etc.)
- Need to focus on opportunities to improve performance and the value chain, rather than only on technologies or component capabilities

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Enablers for fast and effective decision-making

Challenges

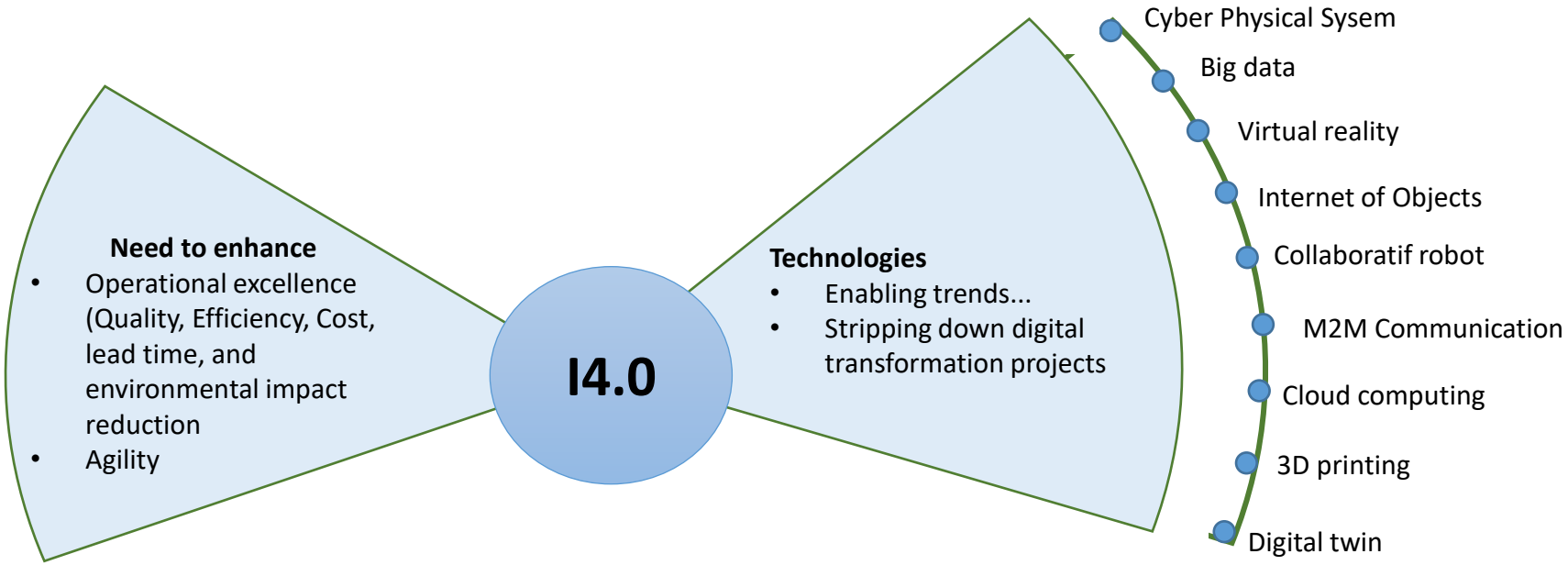


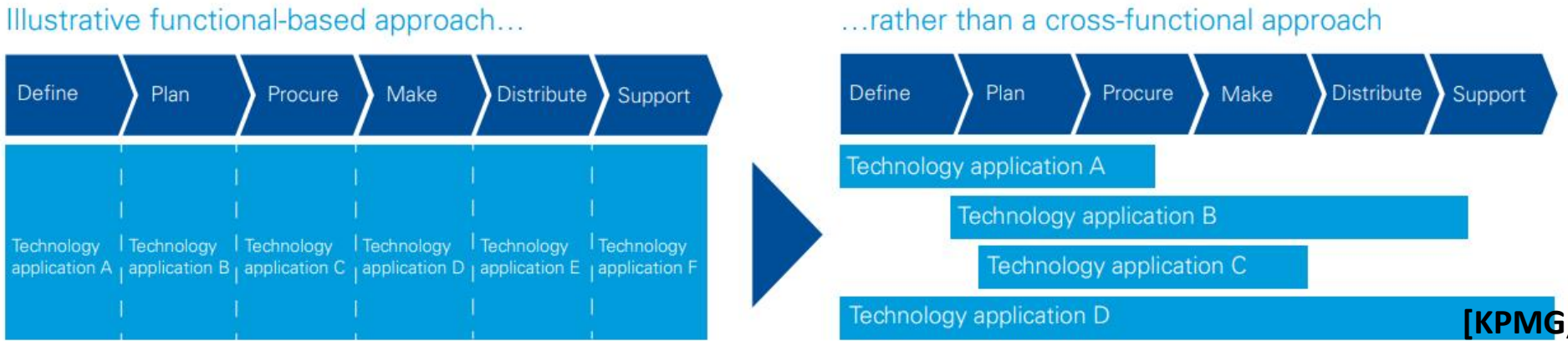


[KPMG, 19]

Enablers for fast and effective decision-making

Challenges





[KPMG, 19]

Contribution of technologies in the decision process

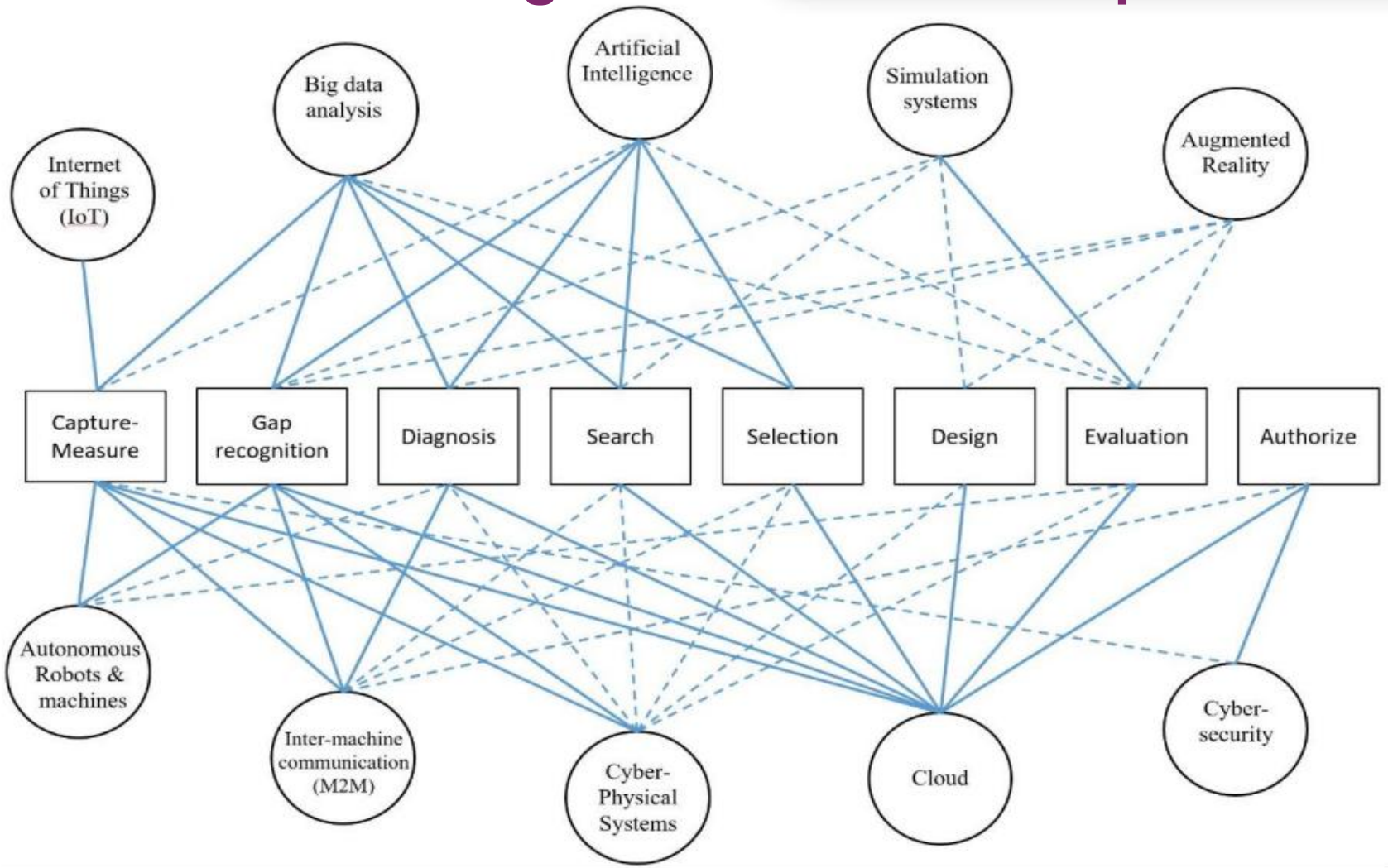


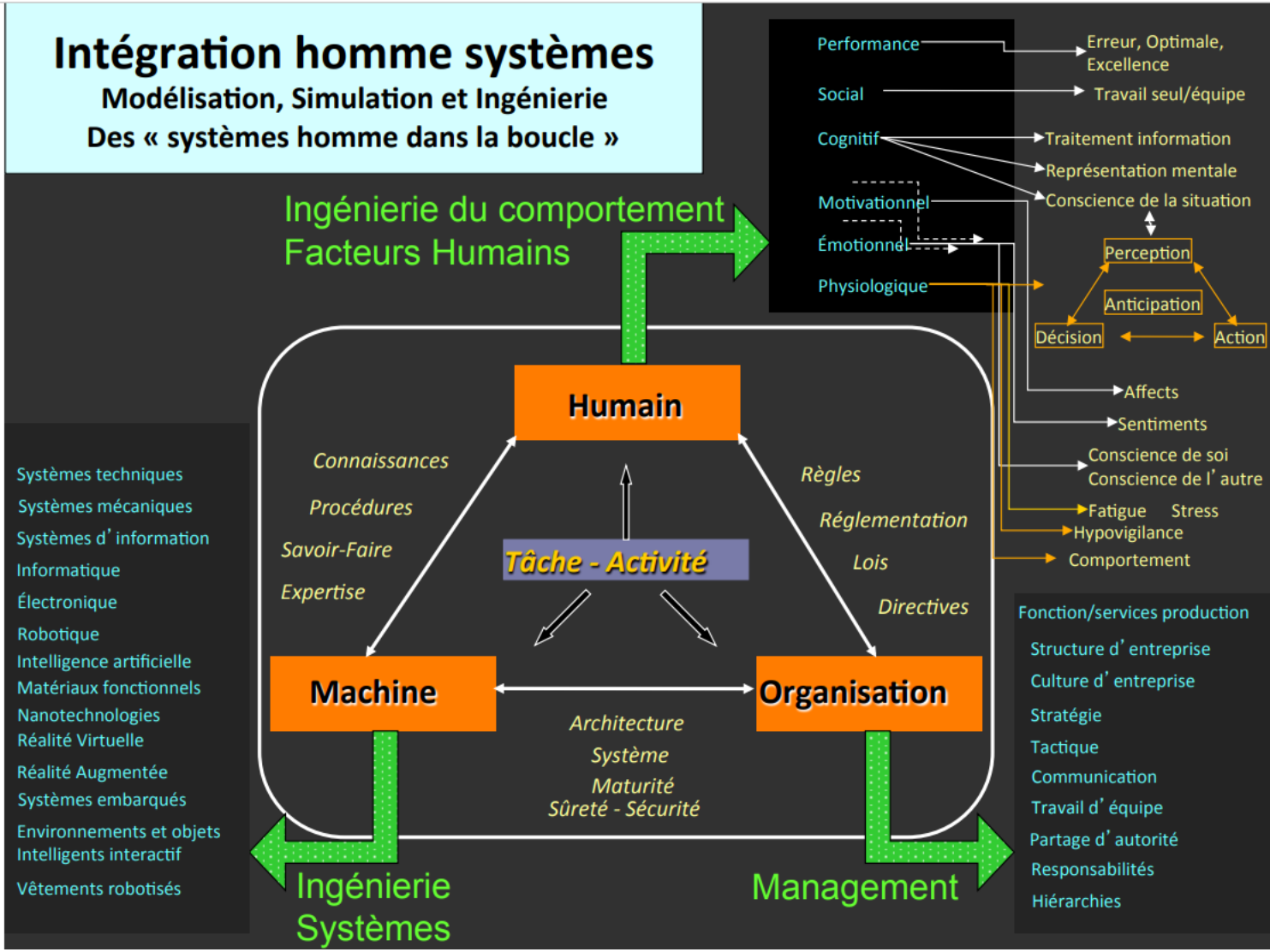
Figure 17 : Contributions of Industry 4.0 technologies to decision-process enhancement that are in consensus or under debate.

F.Rosin Phd thesis 2022

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Human Centred Design consideration

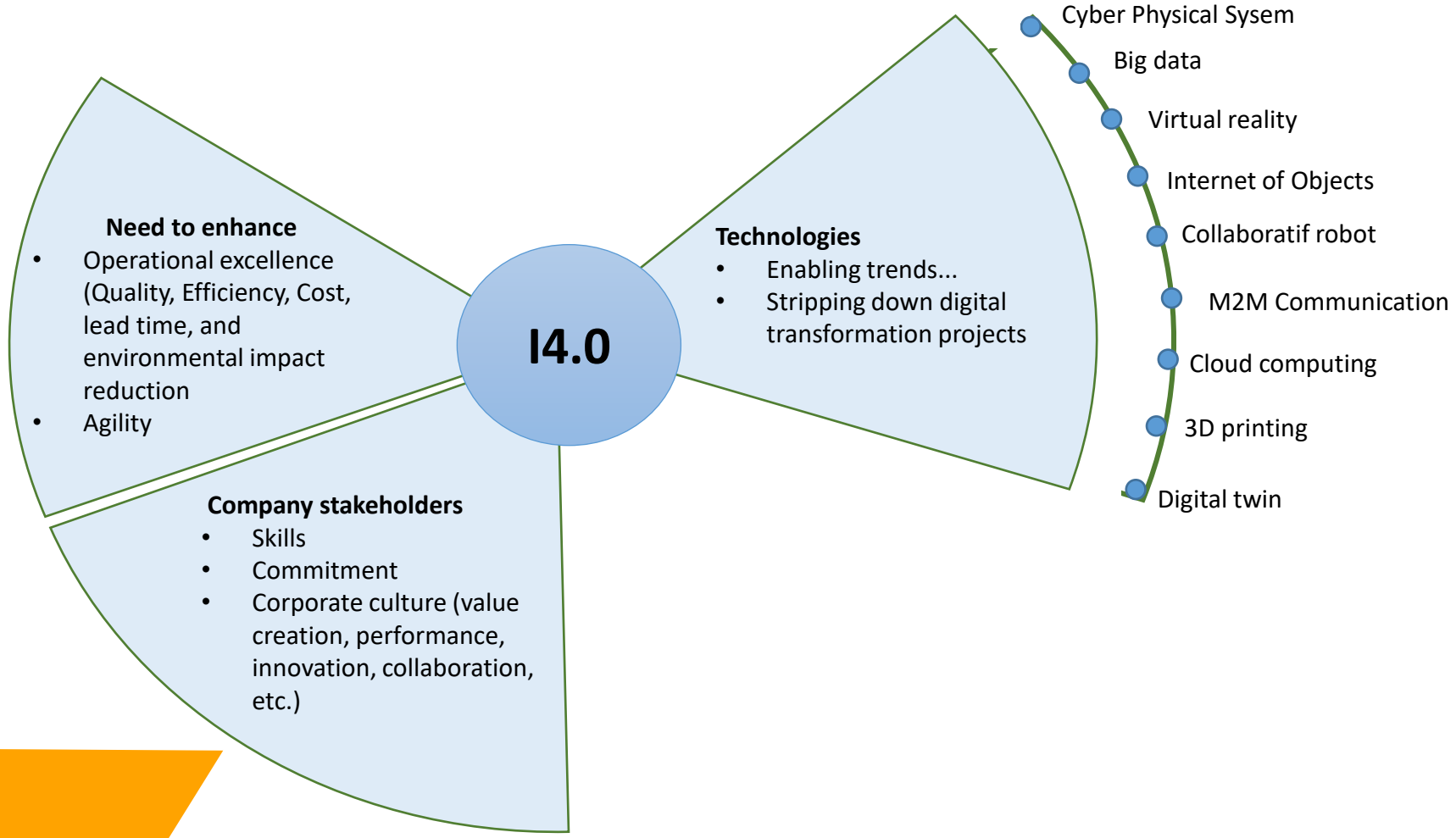
- Develop and validate modeling methods for technical and organizational system design (human–machine)
- Guarantee reliability and strengthen human expertise

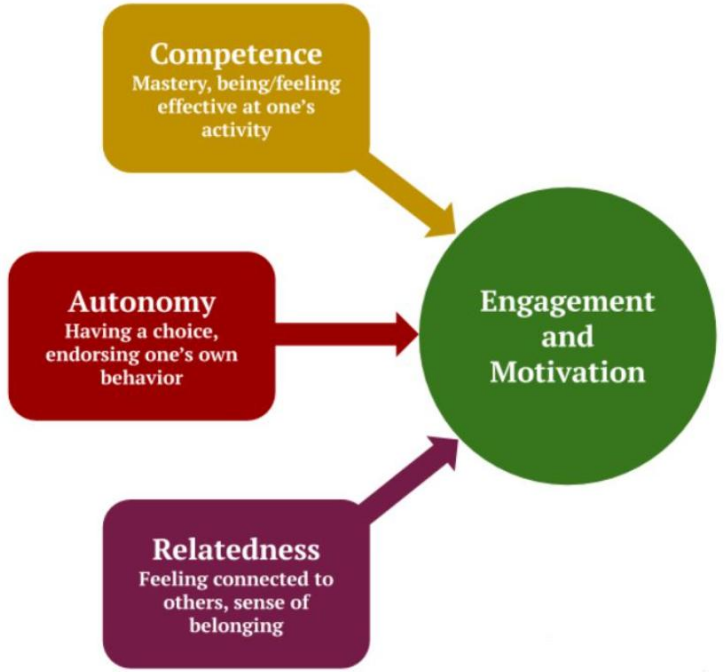


[Fass, 2021]

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Challenges

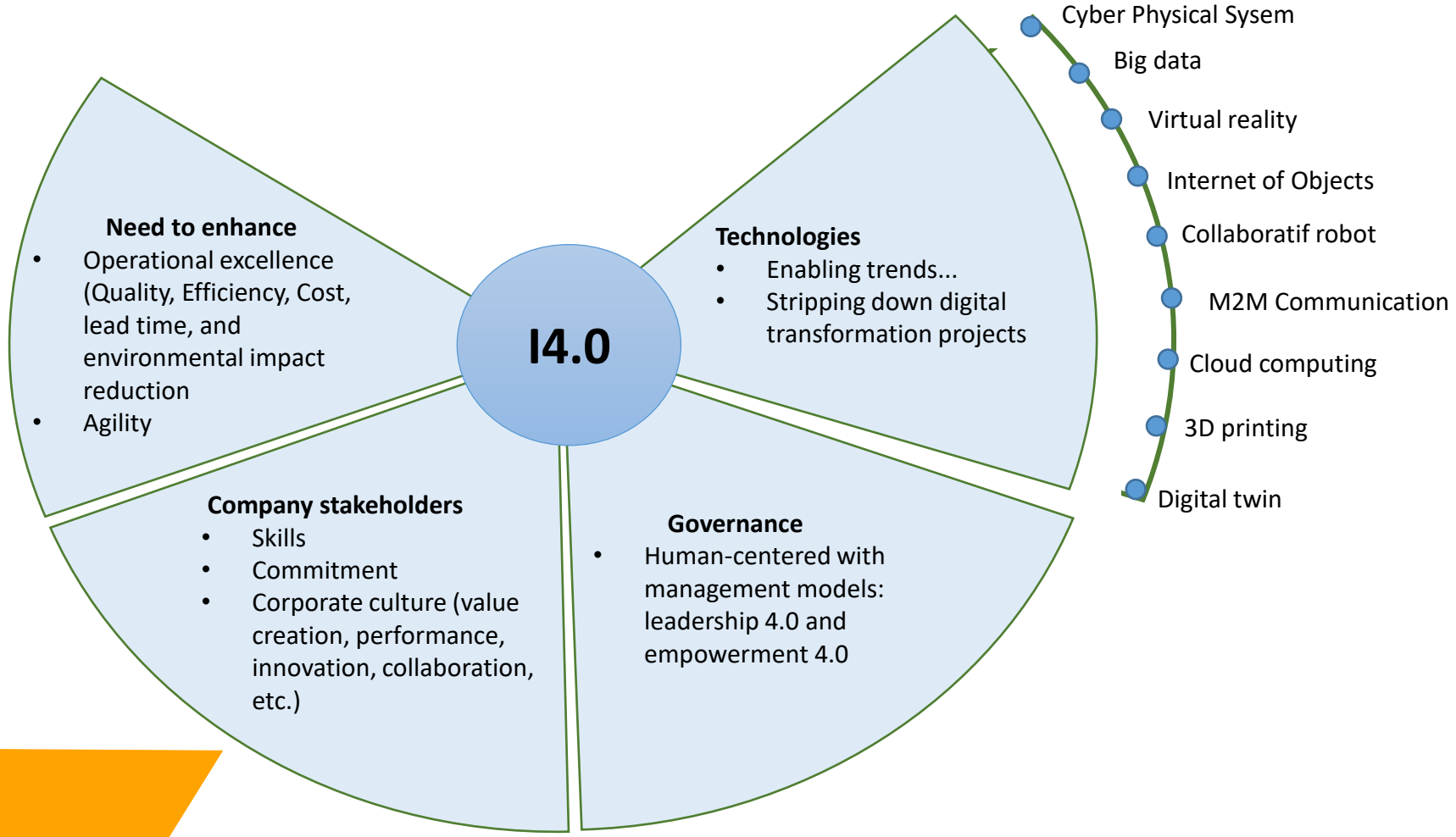


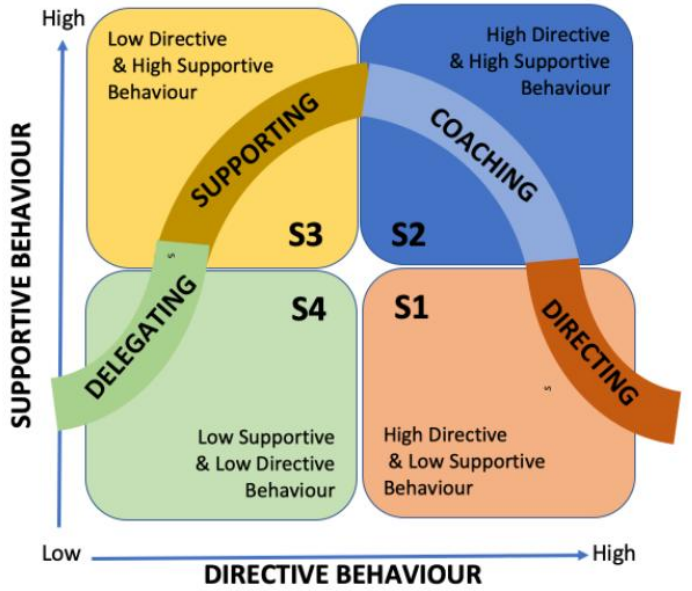


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Enablers for fast and effective decision-making

Challenges



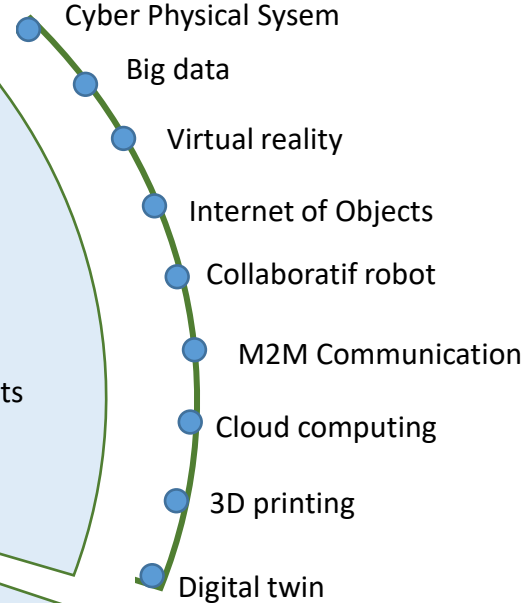


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Enablers for fast and effective decision-making

Challenges

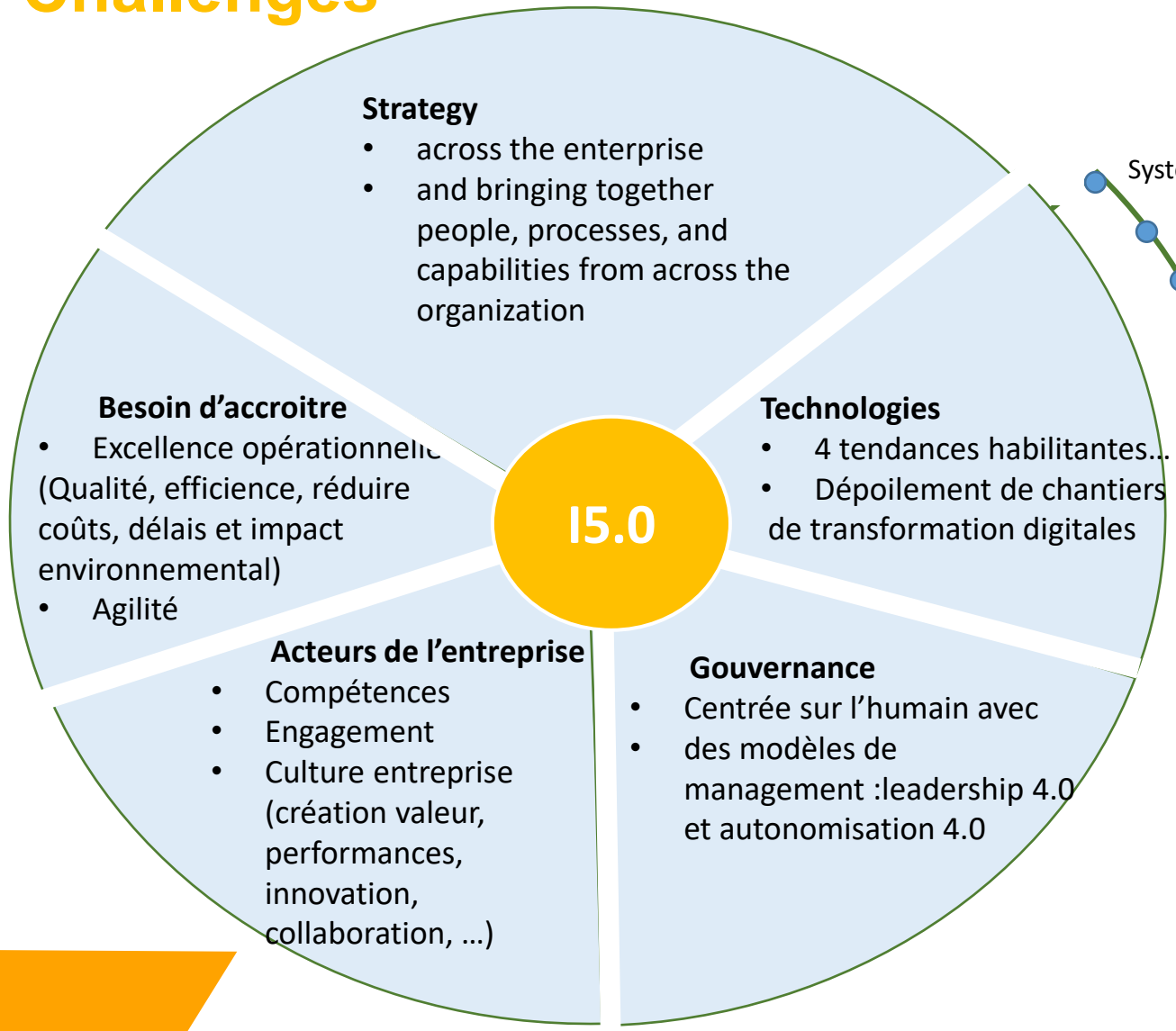


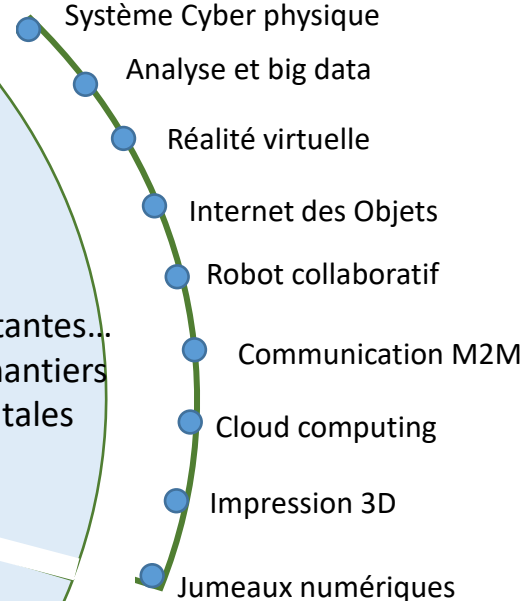


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Enablers for fast and effective decision-making

Challenges

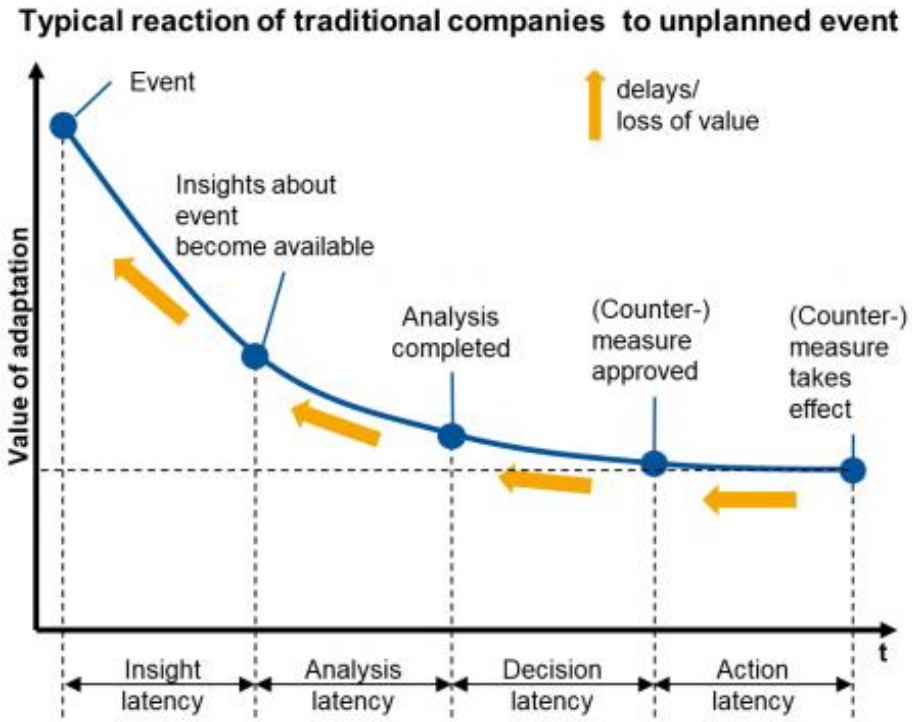




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Analysis of the decision-making process

3.0



- Data related to the workshop, operators, project management, etc. is not reported in real time, except for data from machines, which is reported via PLCs.
- Tracked records and anomalies are recorded on paper and reported as soon as possible.

- If there is a problem, cause/effect analysis cannot be carried out quickly, especially if the relevant person has not been informed in time.
- Not all information related to parameters is available, as there are no instruments for workstations other than automated machine data.
- Information needs to be searched for in different information systems.

Important time to reinvent solutions because there has been no capitalization

Immature decisions based on unreliable or outdated data because the context is dynamic

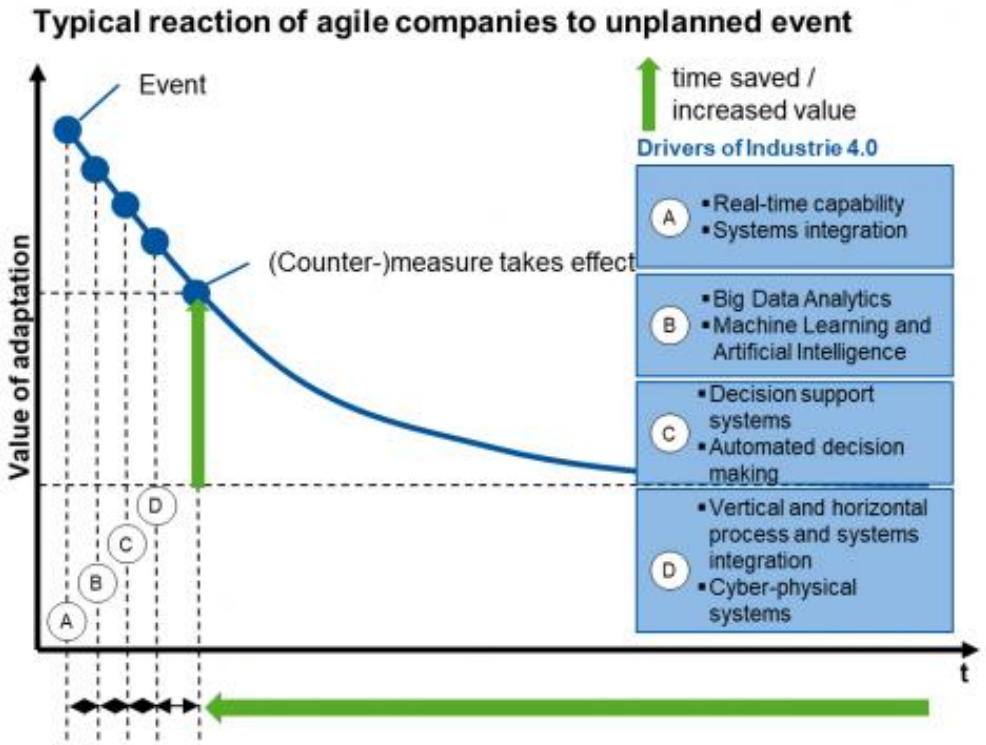
Detecte

Analyse

Decide

Act

Analysis of the decision-making process 4.0



Agility, collaboration, real time

- Data is reported in real time.
- Tracked records and anomalies are digitized, and operators report problems directly.

- If the problem is known because it has been capitalized on in relation to the context parameters that have been reported, the solution is therefore available.
- If the problem is unknown, AI tools can predict possible solutions.
- Simulation tools (such as digital twins) can also be deployed to solve the problem.
- Information systems are interoperable, facilitating data exchange.
- Collaborative problem solving is made easy with visual management tools such as i-obeya.
- A powerful decision support system is deployed and decentralized.

- Mature decisions based on reliable data that take into account changing circumstances
- Towards autonomous decision-making systems

